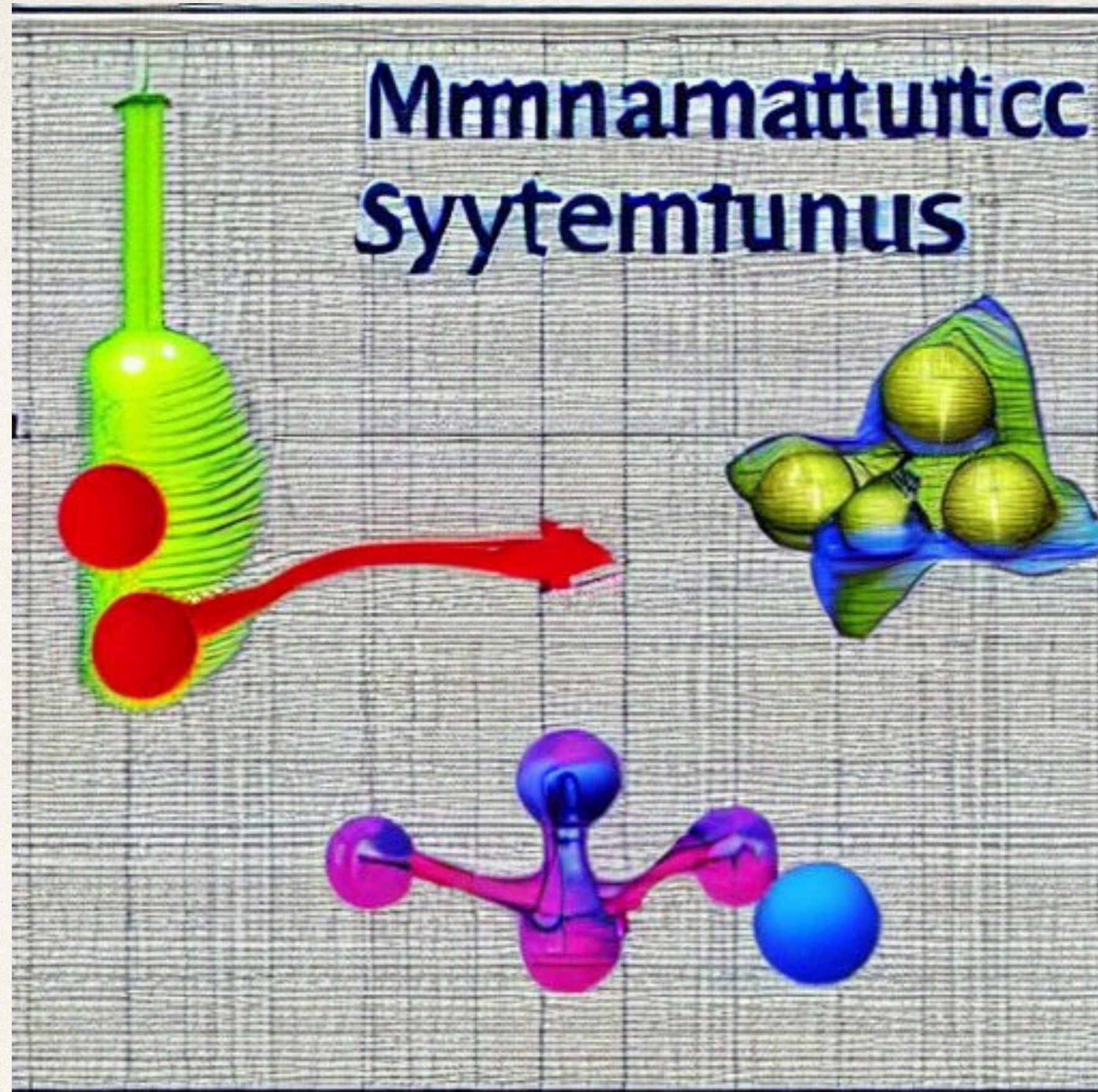




First-principles definitions of agents

Manuel Baltieri

9th Feb 2023

Outline

- ❖ Three classes of first-principles definitions of agents
 - ❖ Prediction-based
 - ❖ Causality-based
 - ❖ Relational
- ❖ Conclusion

Important note

The following frameworks have various goals:

- ❖ Defining individuality
- ❖ Defining autonomy
- ❖ Defining agency
- ❖ ...

I will look at them as characterising agency / agents!

Prediction-based methods

Agency in the eye of the beholder

Prediction-based methods

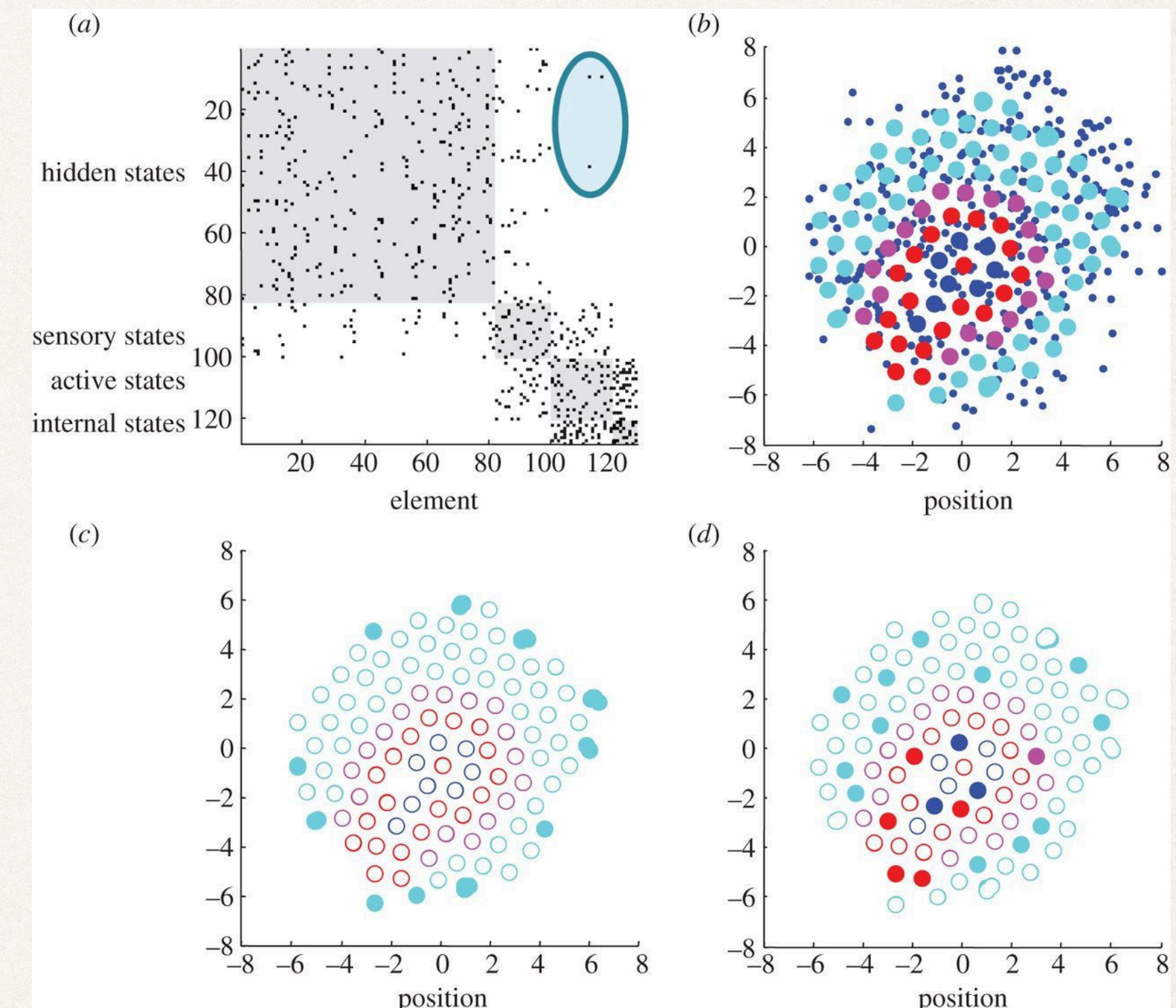
- ❖ **Main idea:** treating a system *AS-IF* it was an agent; this helps predicting its behaviour
- ❖ **Inspiration:** Dennet's intentional stance
- ❖ **Tools:** information theory, filtering theory, Bayesian inference, reinforcement learning, etc.

Examples:

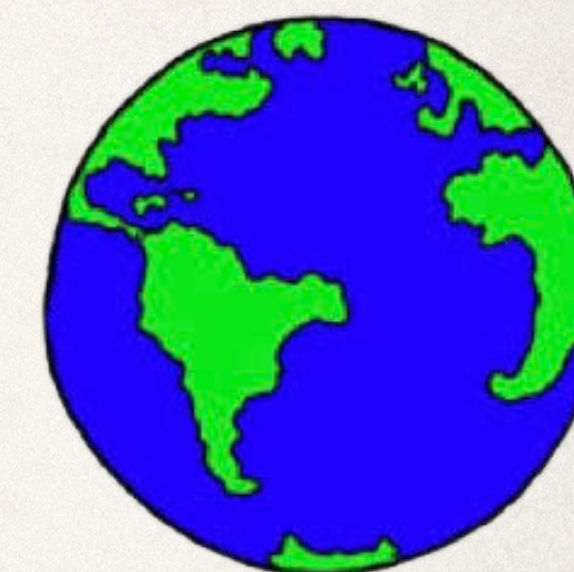
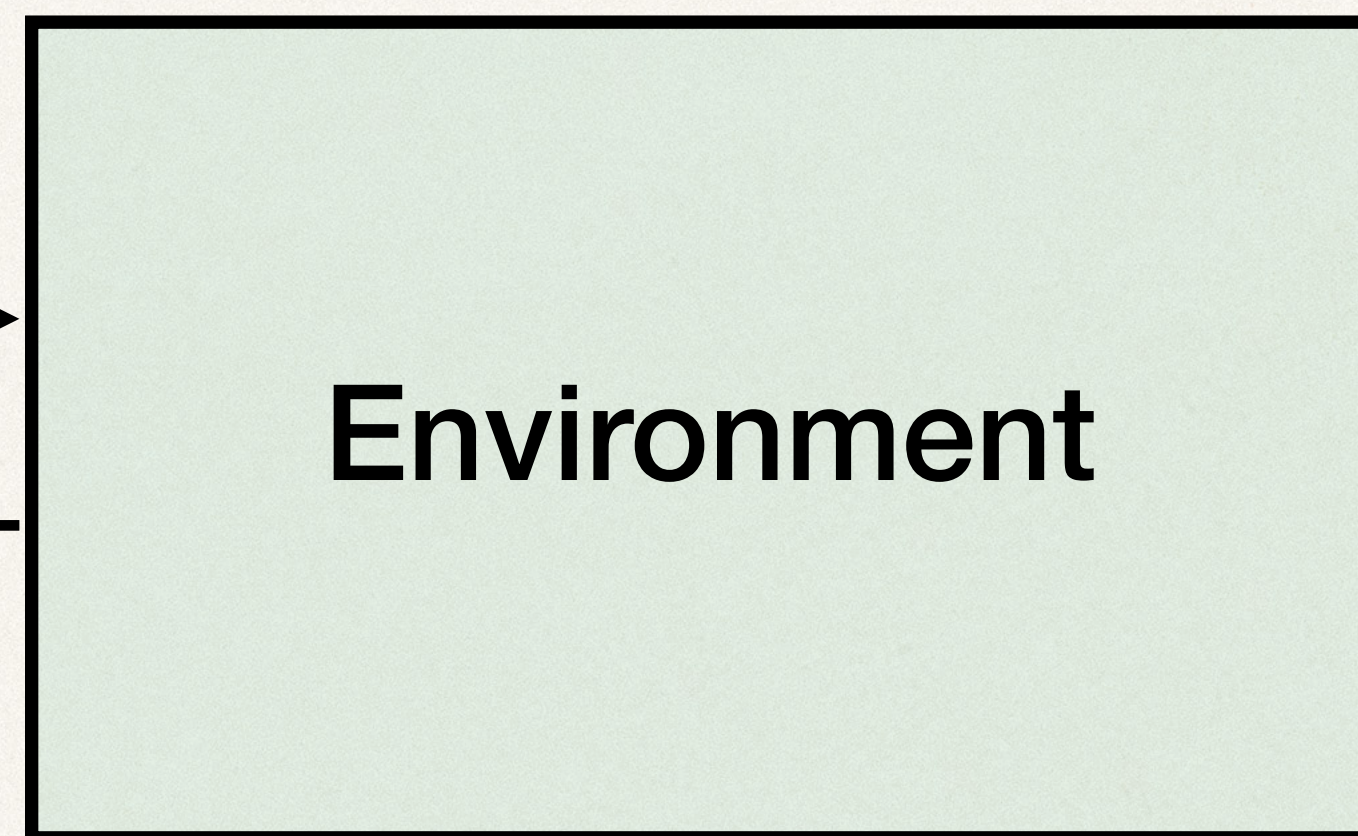
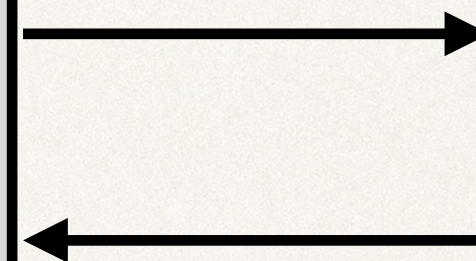
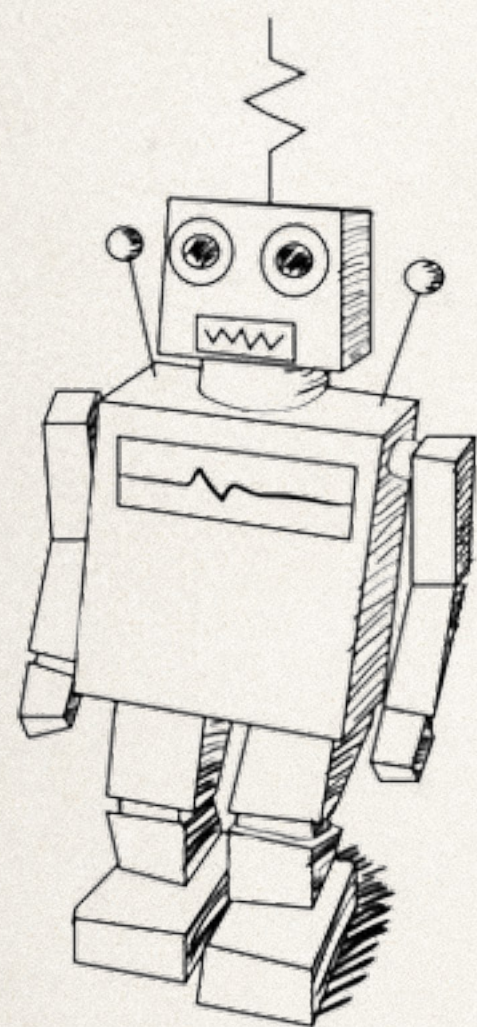
- ❖ The free energy principle
- ❖ The informational individual
- ❖ Behavioural compression

The free energy principle

- ❖ A foundational theory of agents, (living) systems, “things”
- ❖ A thing is a “thing” if and only if it minimises free energy
- ❖ Markov blankets as a veil that separates internal from external states



Friston, K. (2013). Life as we know it. *Journal of the Royal Society Interface*, 10(86), 20130475.



FEP: the agent performs (approximate) Bayesian
inference on the environment

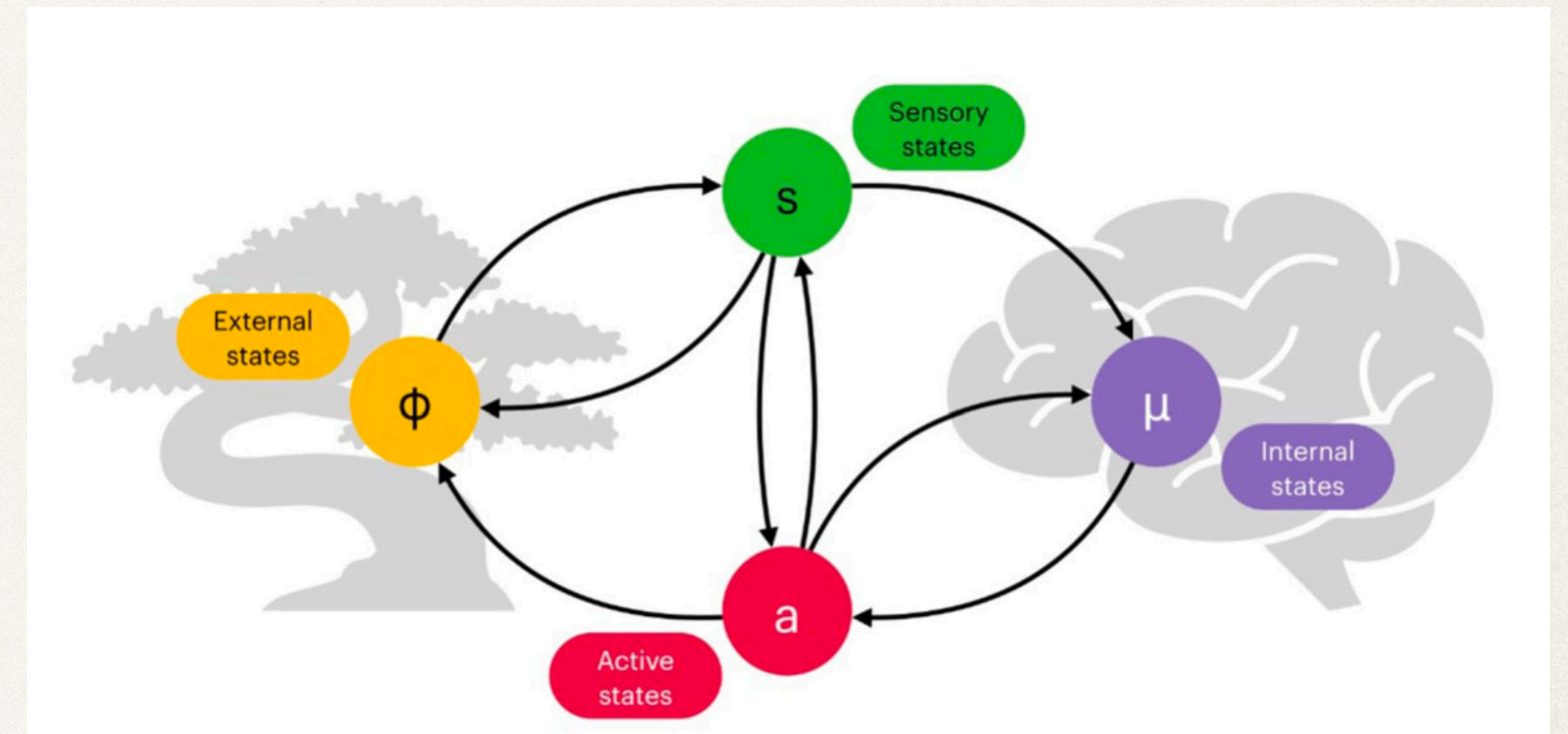
The FEP - pros and cons

Strengths

- ❖ Connections to biology, neuroscience, control theory, reinforcement learning and physics
- ❖ Agency at multiple scales (attempt of finding scale-free theory)

Limitations

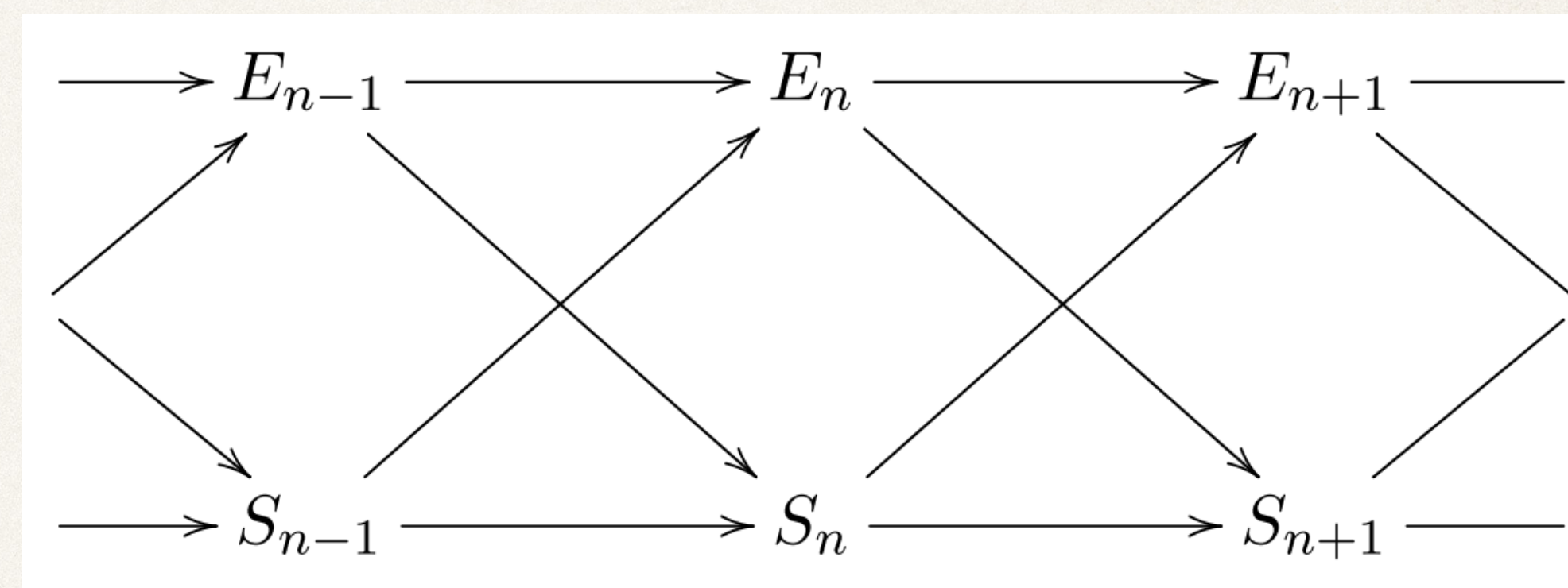
- ❖ Technically limited to stationary processes
- ❖ Ontological commitments (based just on probabilities?)
- ❖ Quantum systems are agents?



Bruineberg, J., Dołęga, K., Dewhurst, J., & Baltieri, M. (2021). The Emperor's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

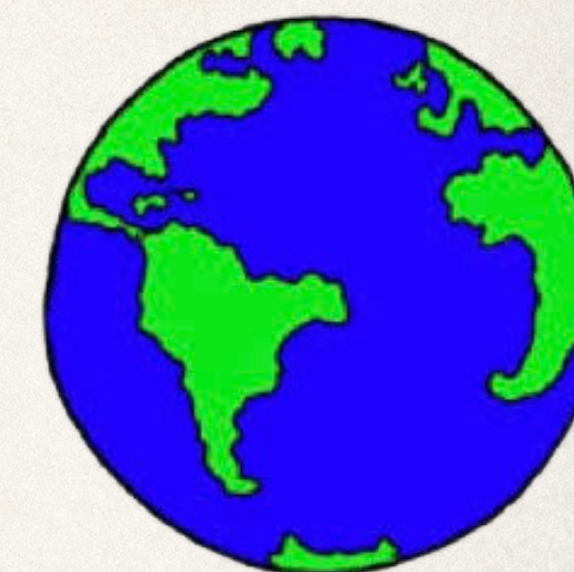
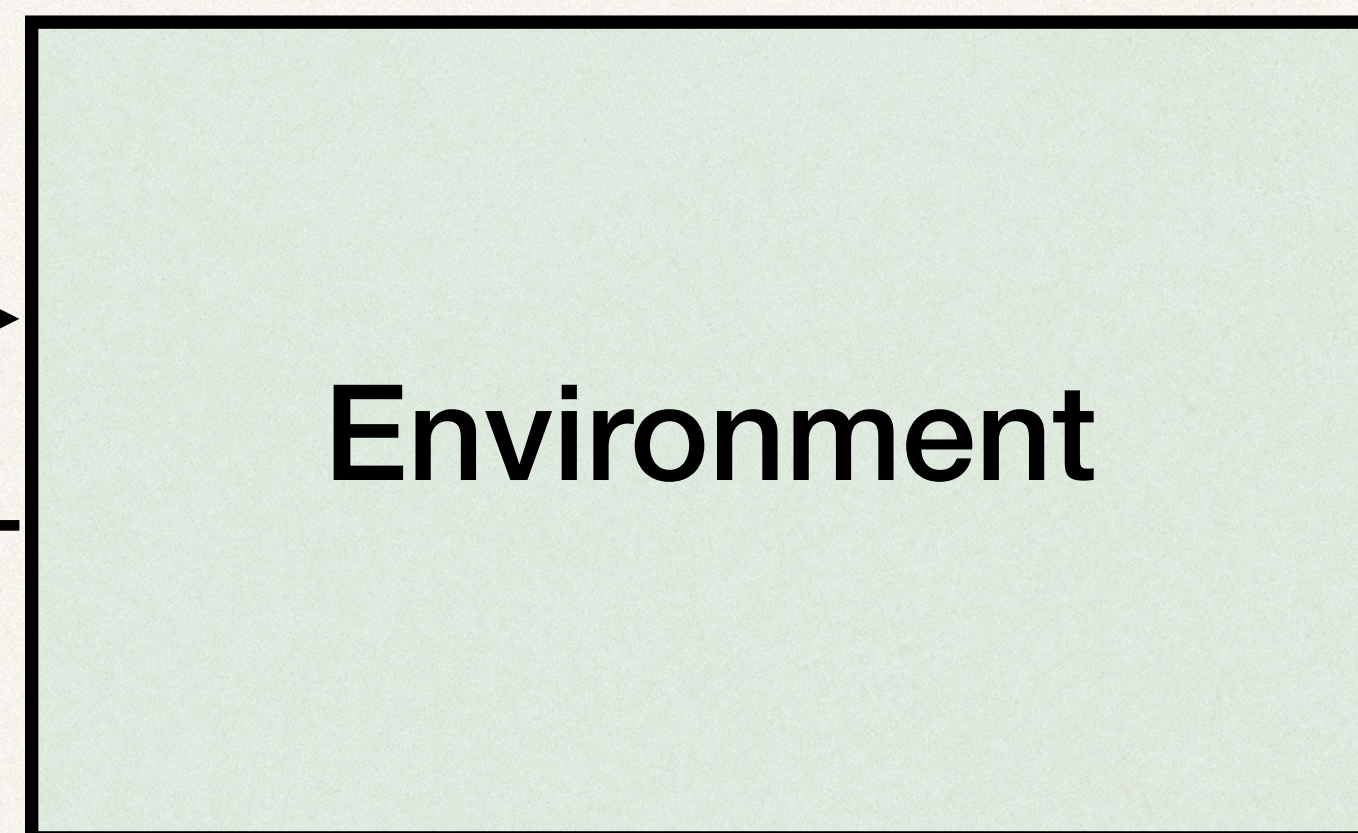
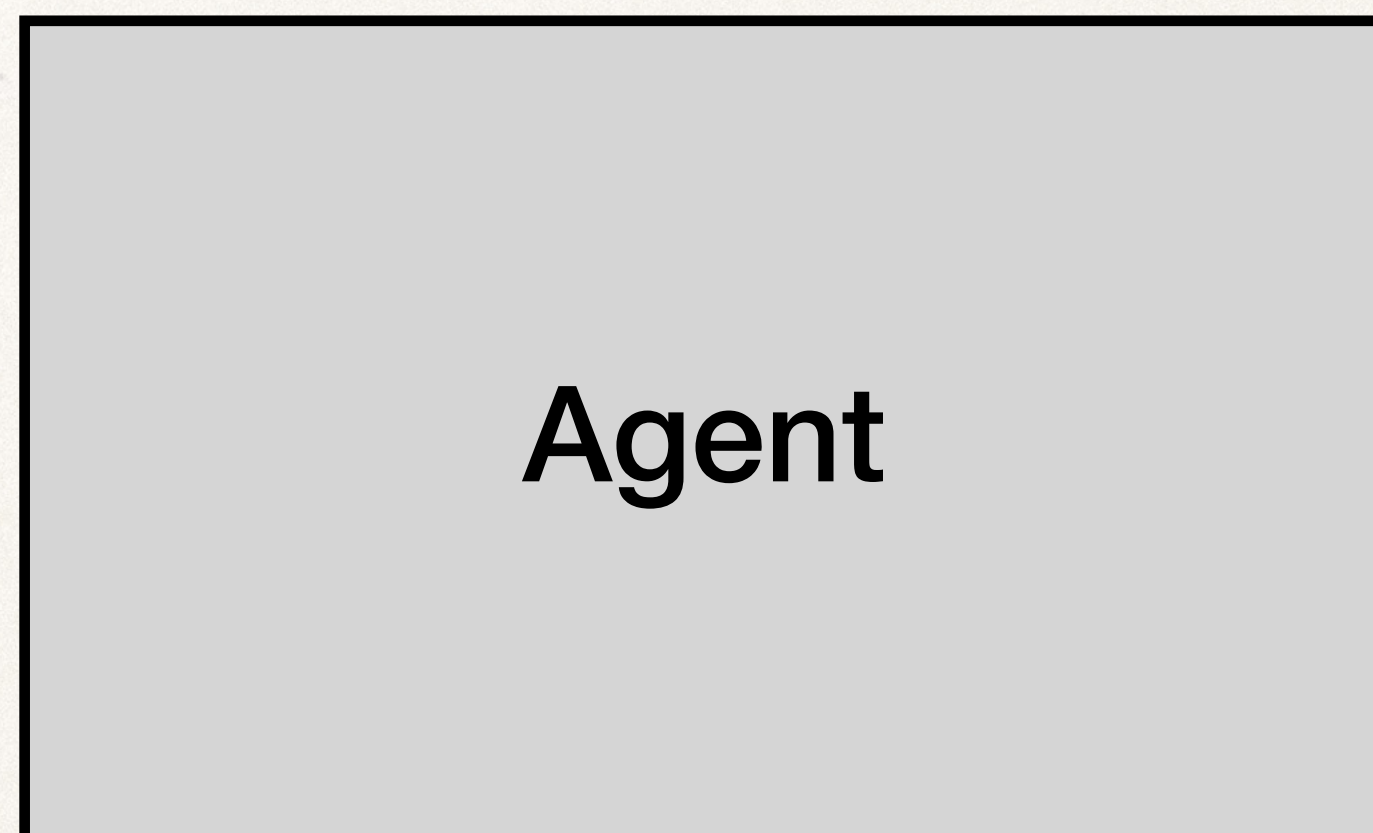
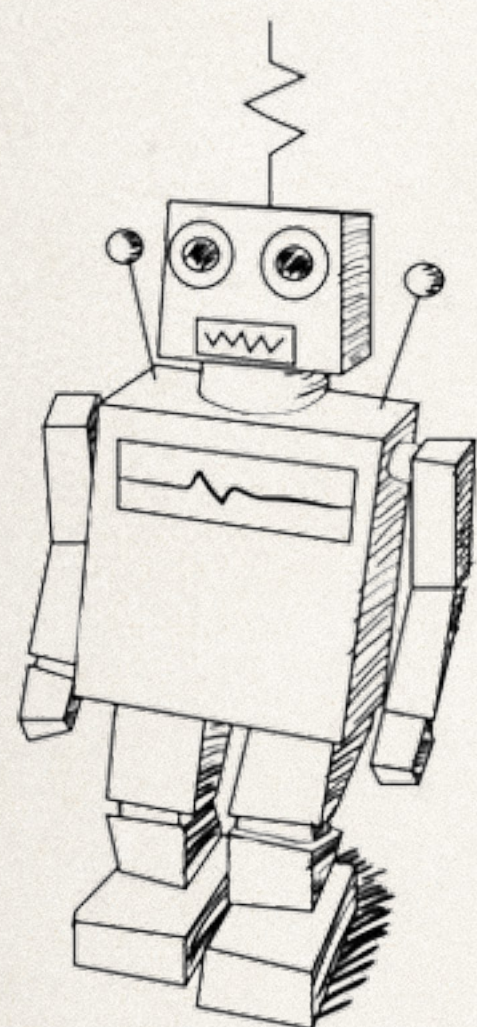
Information individual

- ❖ Information individual: $I(S_n, E_n; S_{n+1})$, how much current state of agent + environment help predict next agent's state
- ❖ First decomposition:
 $I(S_{n+1}; S_n) + I(S_{n+1}, E_n | S_n)$, predictive information of the agent + transfer entropy from environment
- ❖ Second decomposition:
 $I(S_{n+1}; E_n) + I(S_{n+1}, S_n | E_n)$, complementary view



$$\begin{aligned} I(S_n, E_n; S_{n+1}) &= I(S_{n+1}; S_n) + I(S_{n+1}; E_n | S_n) \\ &= I(S_{n+1}; E_n) + I(S_{n+1}; S_n | E_n) \end{aligned}$$

Krakauer, D., Bertschinger, N., Olbrich, E., Flack, J. C., & Ay, N. (2020). The information theory of individuality. *Theory in Biosciences*, 139, 209-223.



Information individual: an agent is good at predicting its future self

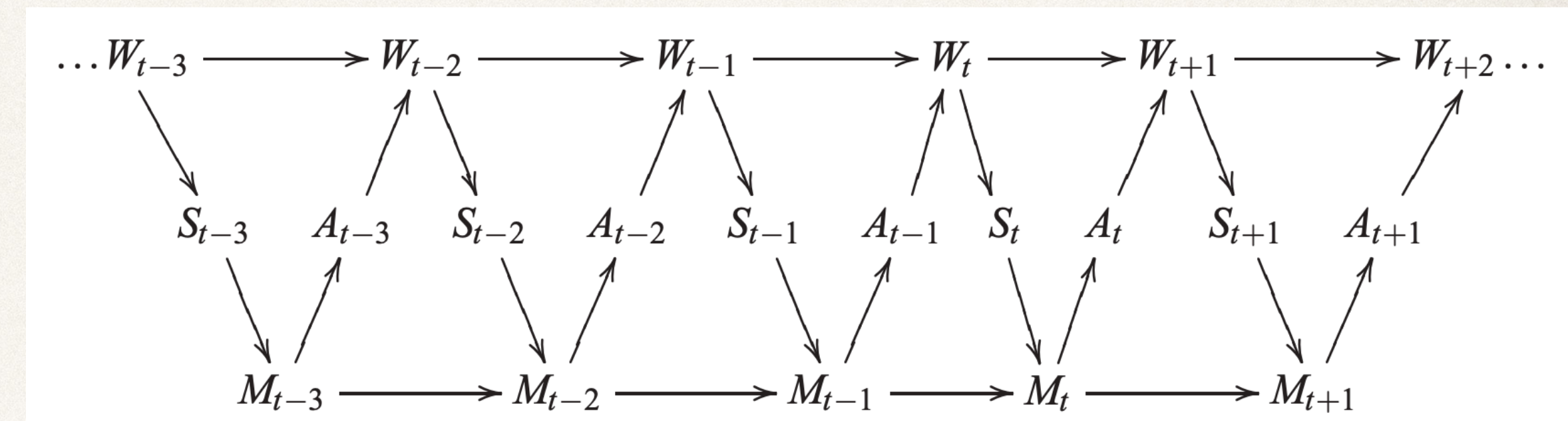
Information individual - pros and cons

Strengths

- ❖ No assumptions about the underlying physics
- ❖ Multi-level agency (group, individual, etc.)
- ❖ Challenging the role of boundaries (e.g., cell membrane)

Limitations

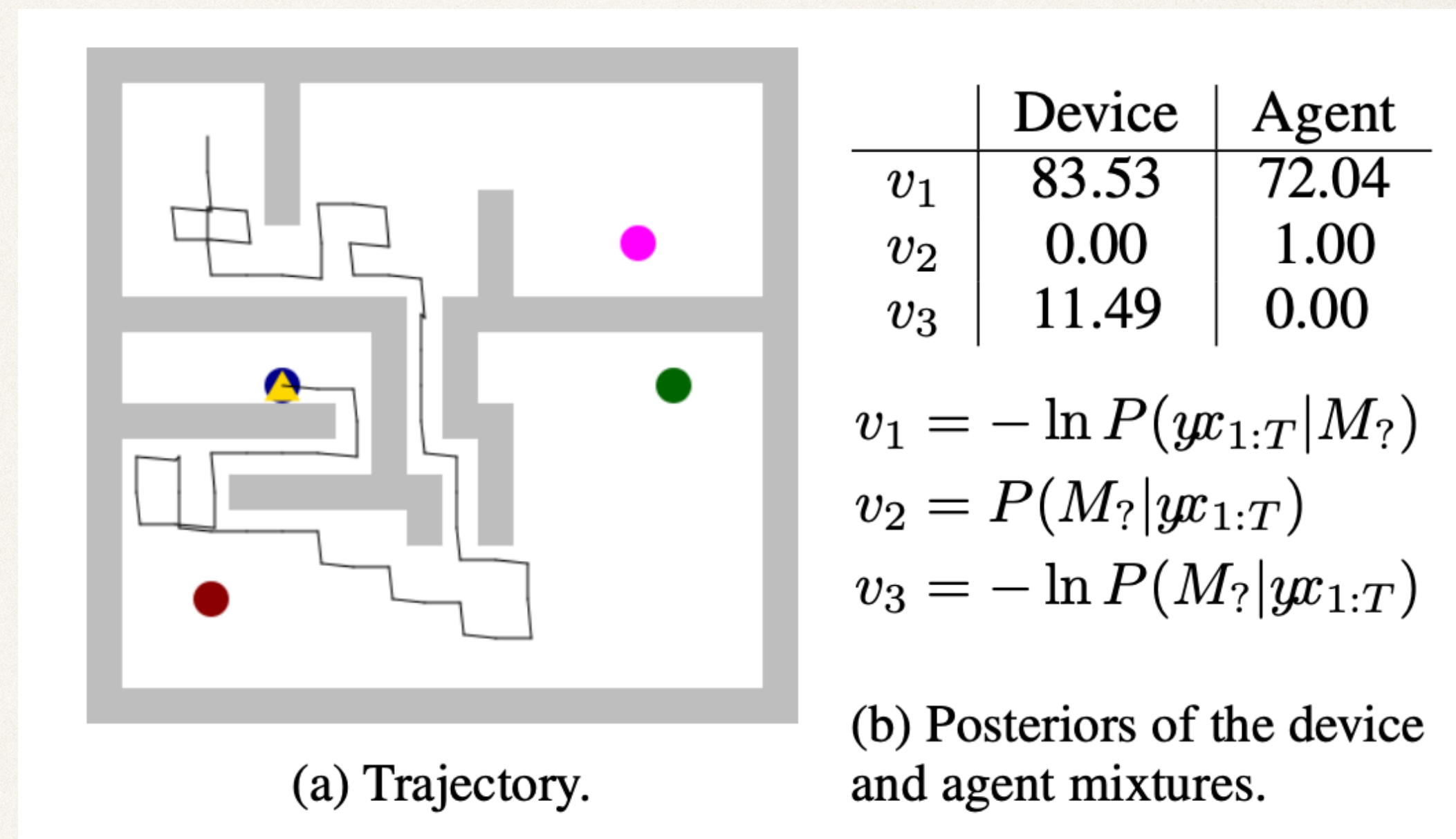
- ❖ No clear rule to generate agent-environment partitions
- ❖ No action
- ❖ Only for discrete time systems



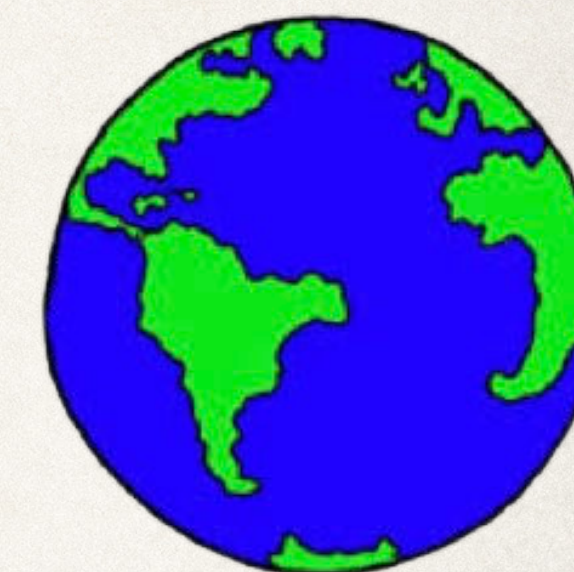
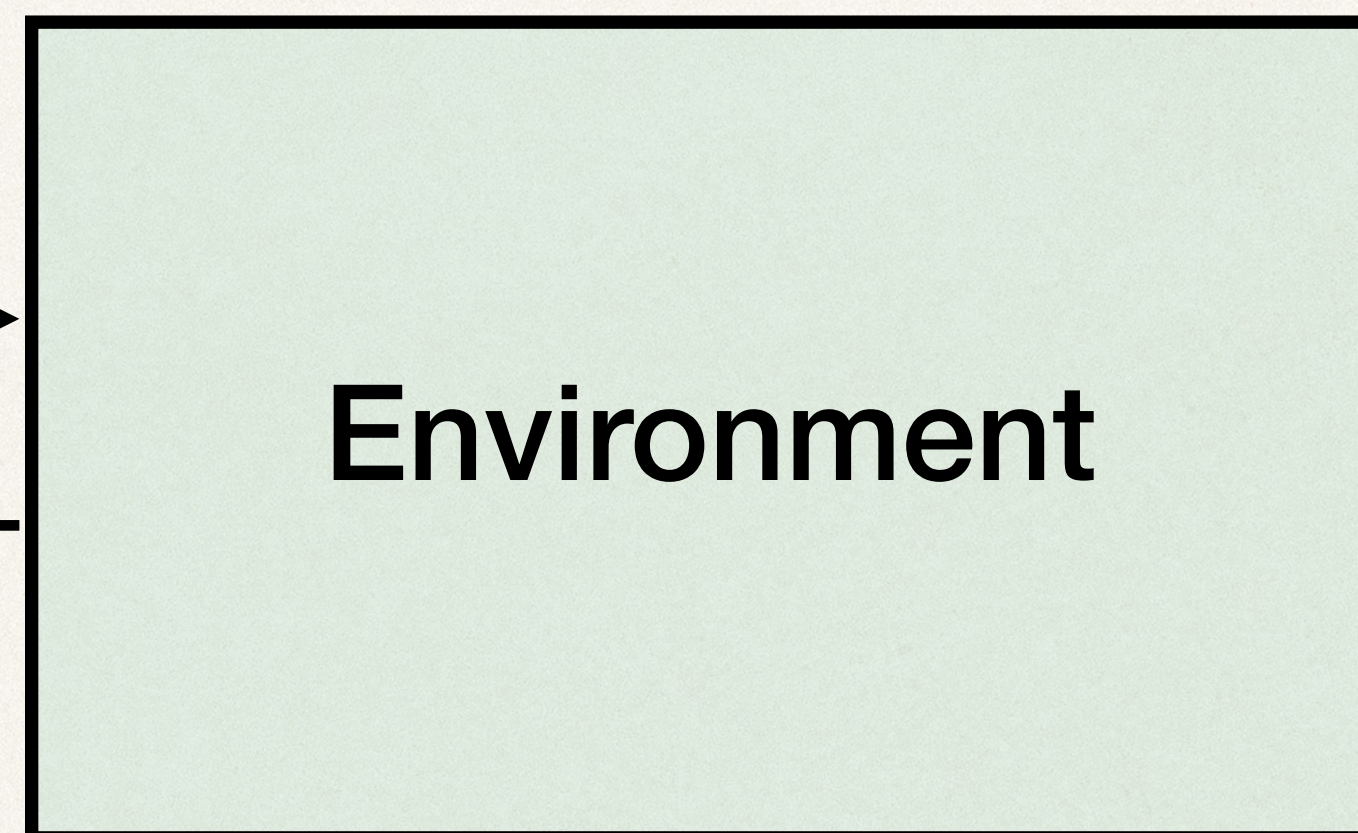
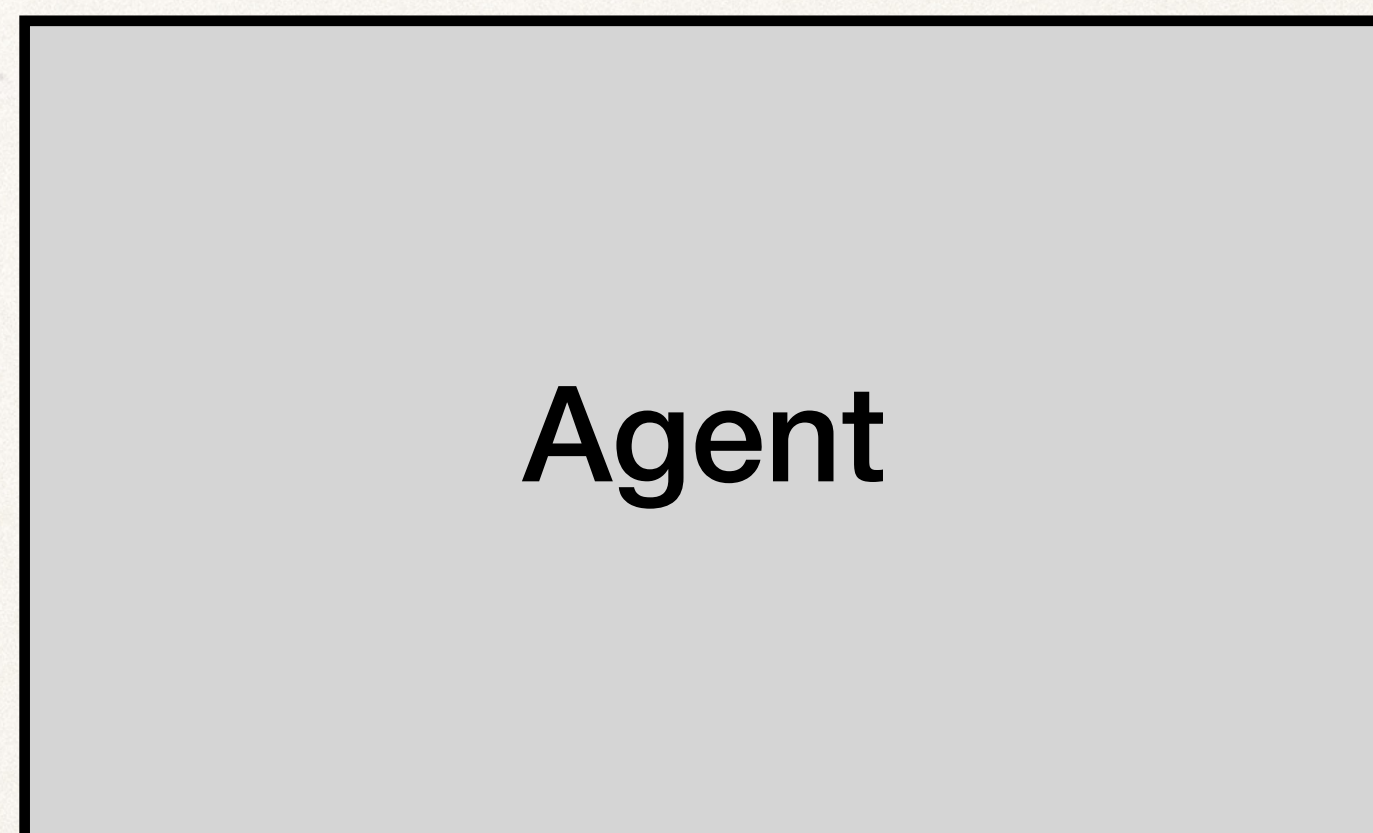
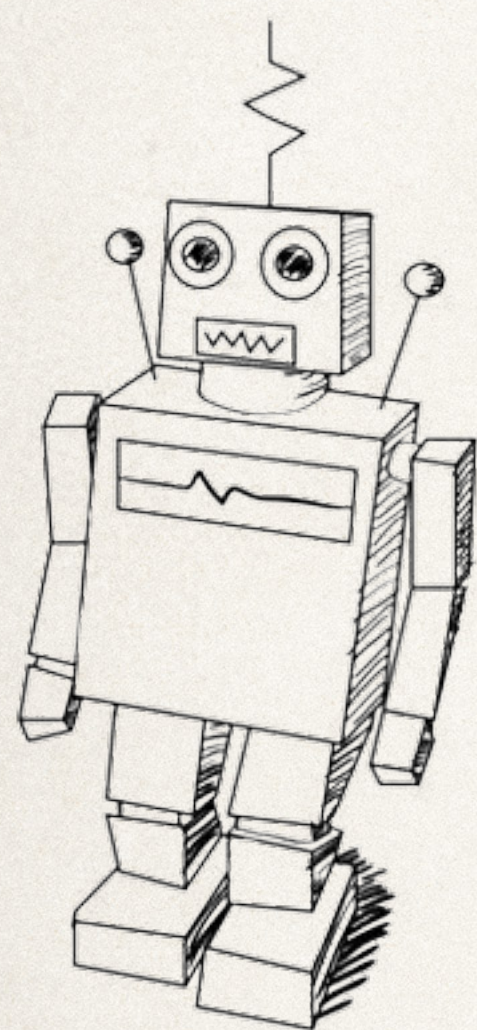
Tishby, N., & Polani, D. (2011). Information theory of decisions and actions. Perception-action cycle: Models, architectures, and hardware, 601-636.

Behavioural compression

- ❖ A formalisation of the intentional stance (cf., a stone, a thermostat and a game-playing computer)
- ❖ Use inverse reinforcement learning to find the best possible goal for a system
- ❖ Use methods from algorithmic probability to find the simplest description of a trajectory of a system (no goals)
- ❖ Compare RL agents with policies for planning to reactive systems with step-by-step predictions



Orseau, L., McGill, S. M., & Legg, S. (2018). Agents and devices: A relative definition of agency. arXiv preprint arXiv:1805.12387.



Behavioural compression: goals help predicting the long-term
behaviour of a system

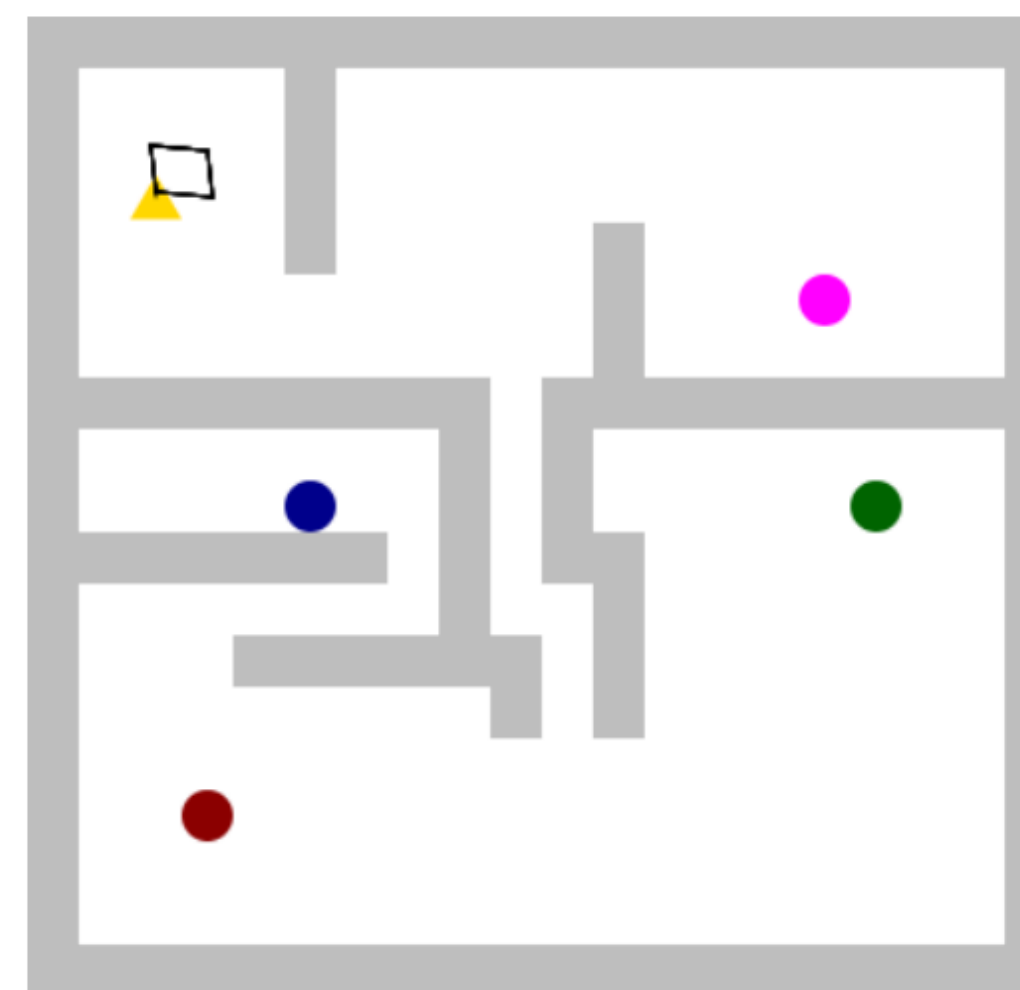
Behavioural compression - pros and cons

Strengths

- ❖ Formalising intentional stance
- ❖ Allows to consider different goals and compression strategies

Limitations

- ❖ Not clear what happens when behaviour can't be compressed
- ❖ Can't discriminate between agents, and systems behaving AS IF they were agents
- ❖ Unclear whether different choices (goals, predictors) would influence the final results



(a) Trajectory.

	Device	Agent
v_1	18.01	37.48
v_2	1.00	0.00
v_3	0.00	19.40

$$v_1 = -\ln P(yx_{1:T}|M_?)$$

$$v_2 = P(M_?|yx_{1:T})$$

$$v_3 = -\ln P(M_?|yx_{1:T})$$

(b) Posteriors of the device and agent mixtures.

Orseau, L., McGill, S. M., & Legg, S. (2018). Agents and devices: A relative definition of agency. arXiv preprint arXiv:1805.12387.

Prediction-based methods

Advantages

- ❖ They take into account the role of observers
- ❖ Agnostic about the underlying system (we just need some notion of information)
- ❖ Generally flexible enough to consider multiple scales

Disadvantages

- ❖ Observer-dependent measures (agency is nothing else?)
- ❖ Different existing measures of information
- ❖ Can't distinguish between “real” and “as-if” agents

Causality-based methods

Agency as a property intrinsic to a system

Causality-based methods

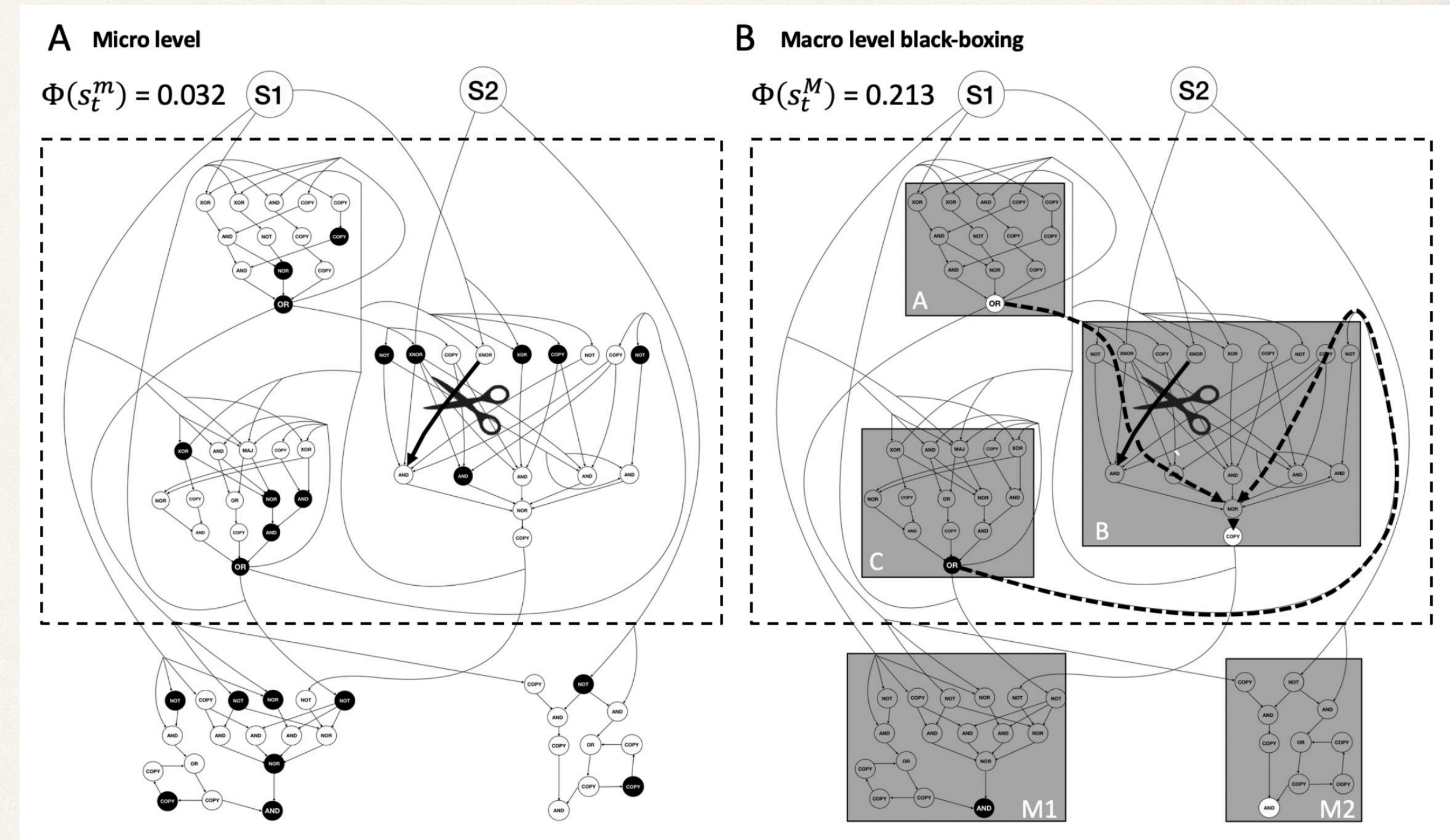
- ❖ **Main idea:** actions are causal
- ❖ **Inspiration:** Davidson's "causalism", mental states cause actions in the world; Pearl causality
- ❖ **Tools:** do calculus, information theory, Bayesian networks, etc.

Examples:

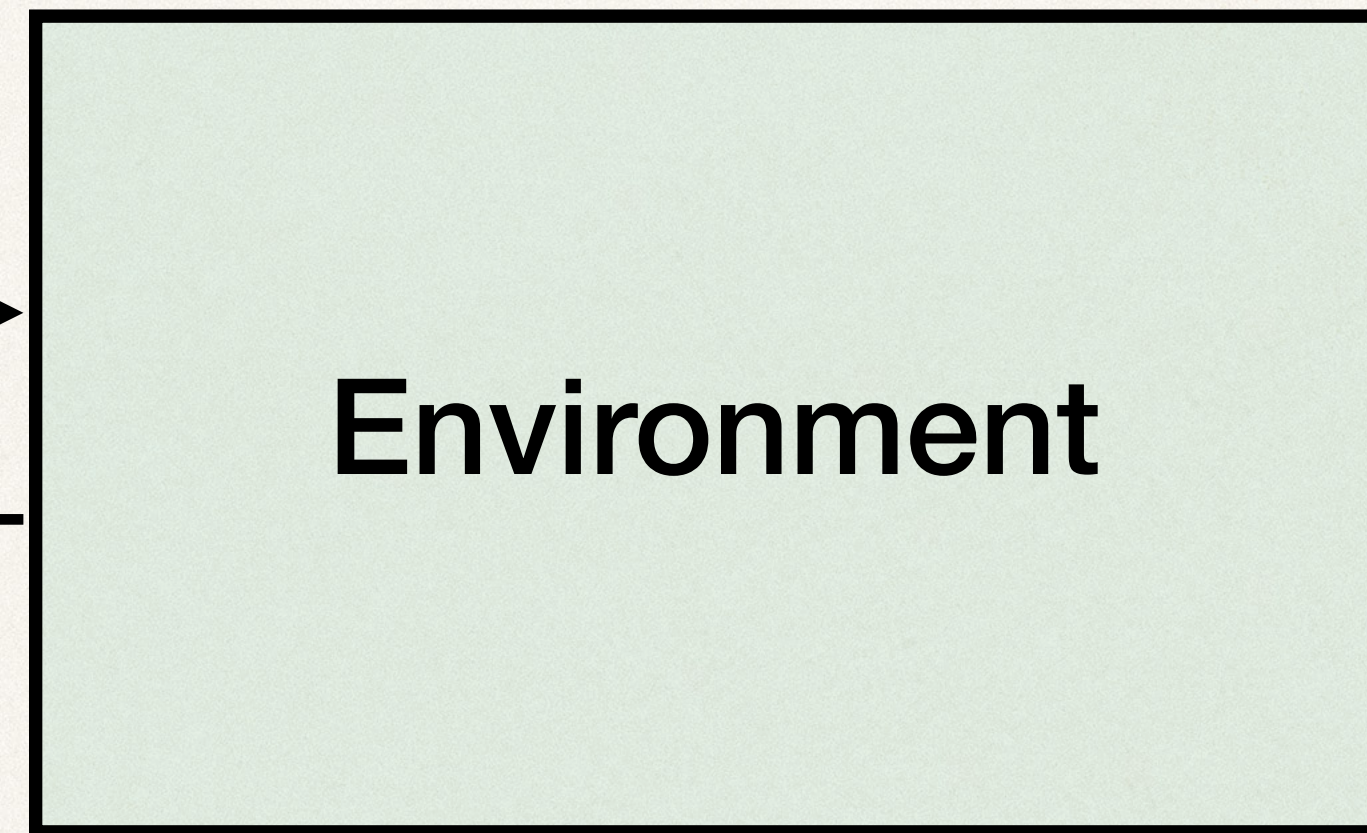
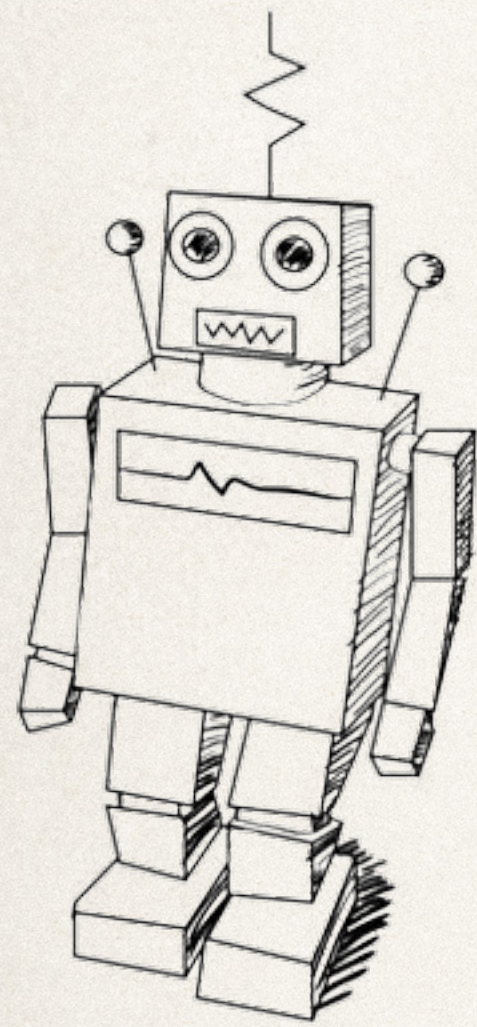
- ❖ Integrated information theory
- ❖ Semantic information
- ❖ Mechanised causal graphs

Integrated information theory

- ❖ A foundational theory of consciousness (originally), also used for agency recently
- ❖ IIT quantifies the “intrinsic irreducibility” of a system: how much cause-effect power of the whole cannot be reduced to its parts
- ❖ The highest intrinsic irreducibility of all possible levels determines the system of interest (conscious system, agent, etc.)



Albantakis, L., Massari, F., Beheler-Amass, M., & Tononi, G. (2021).
A macro agent and its actions. In *Top-Down Causation and Emergence* (pp. 135-155). Cham: Springer International Publishing.



IIT: agents are emergent, they can't be explained
with a reductionist approach

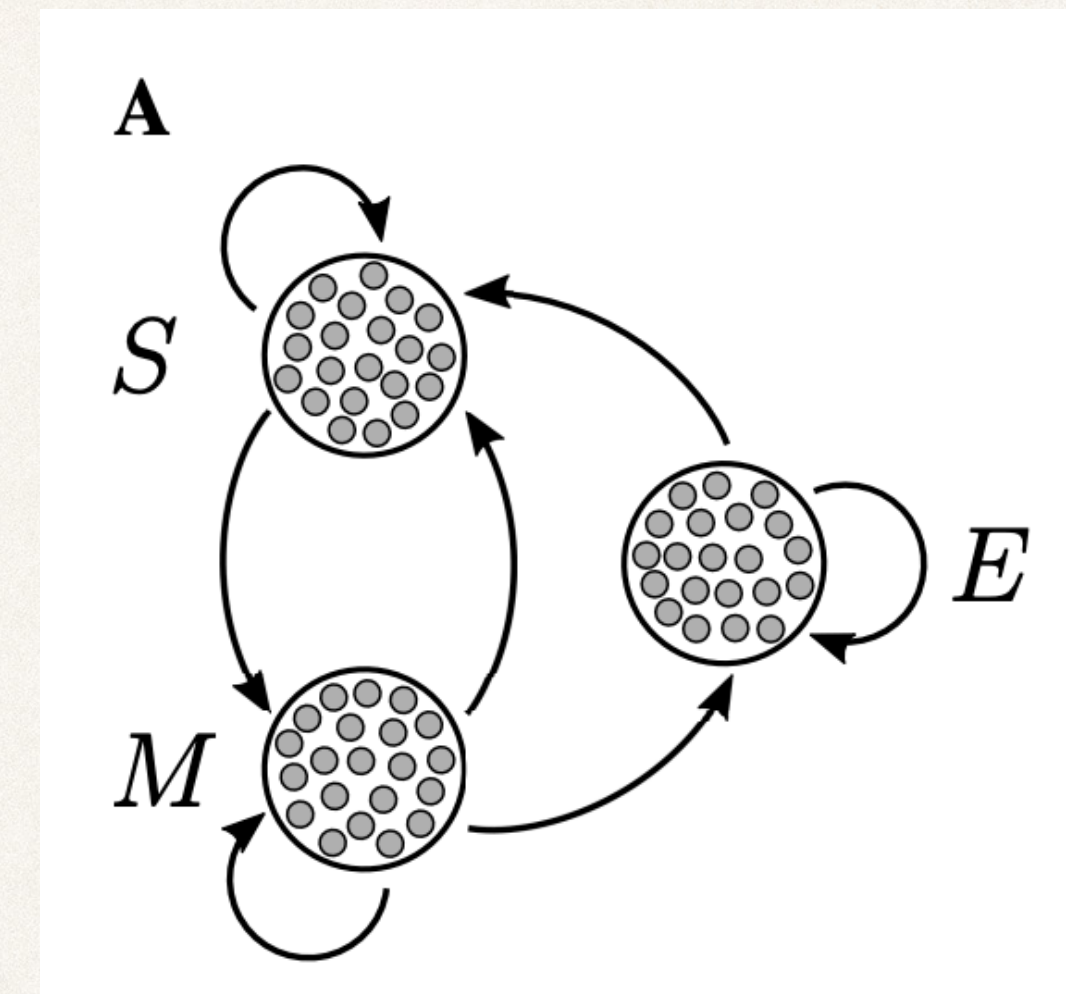
IIT - pros and cons

Strengths

- ❖ Formalising agency discovery at multiple scales
- ❖ Formalising a way to consider emergence vs. reductionist explanations
- ❖ Causal vs observational analysis

Limitations

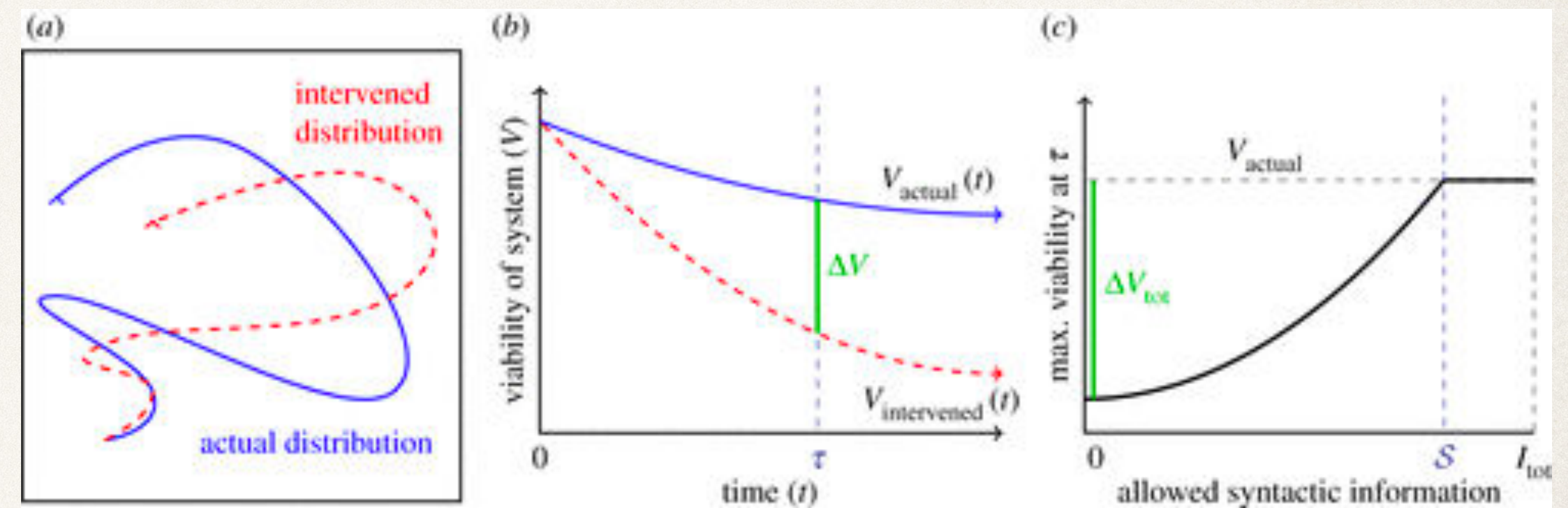
- ❖ Agency (consciousness) only defined at a single scale
- ❖ General, potential issues with IIT (4.0 version just came out to fix some, or add more?)
- ❖ Only for discrete time systems



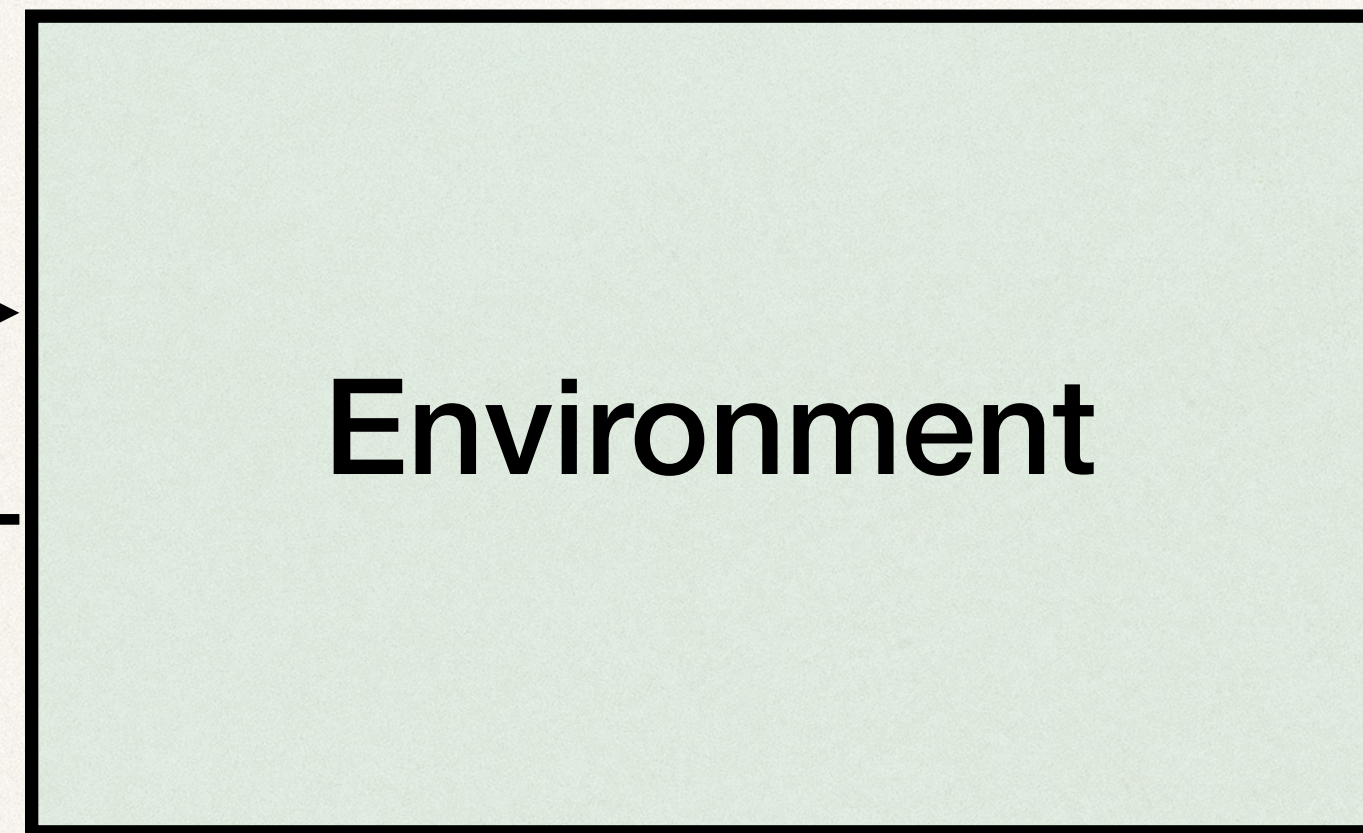
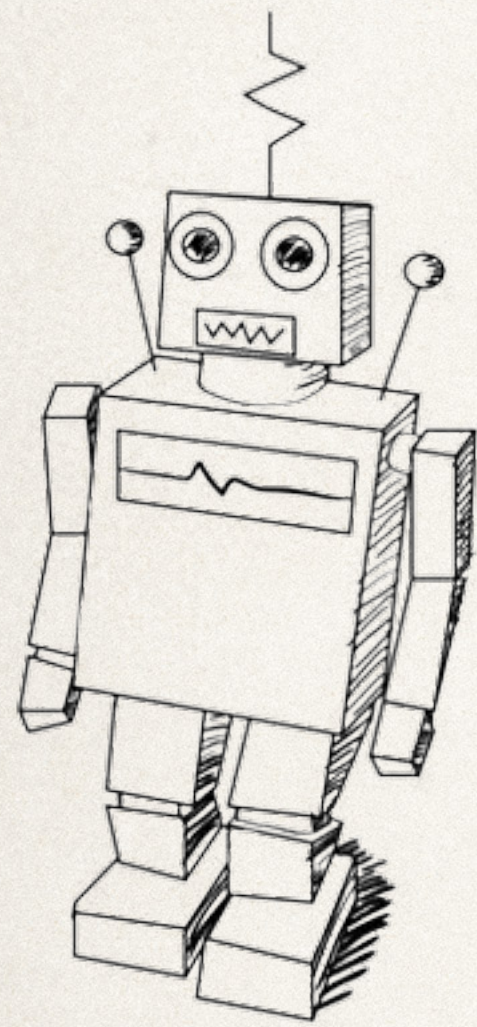
Aguilera, Miguel, and Ezequiel A. Di Paolo.
"Integrated information and autonomy in the
thermodynamic limit." ALIFE 2018: The 2018
Conference on Artificial Life. MIT Press, 2018.

Semantic information

- ❖ Shannon information captures correlations, cannot be used to quantify value (semantics)
- ❖ Semantic information is a way to provide non-correlational (“scrambled”) proxies of standard information measures after choosing a particular goal (e.g., survival)
- ❖ Store semantic information: “scrambled” mutual information between agent and environment
- ❖ Observed semantic information: “scrambled” transfer entropy between environment and agent



Kolchinsky, A., & Wolpert, D. H. (2018). Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface focus*, 8(6), 20180041.



Semantic info: agents are systems with a high degree of stored semantic information and observed semantic information

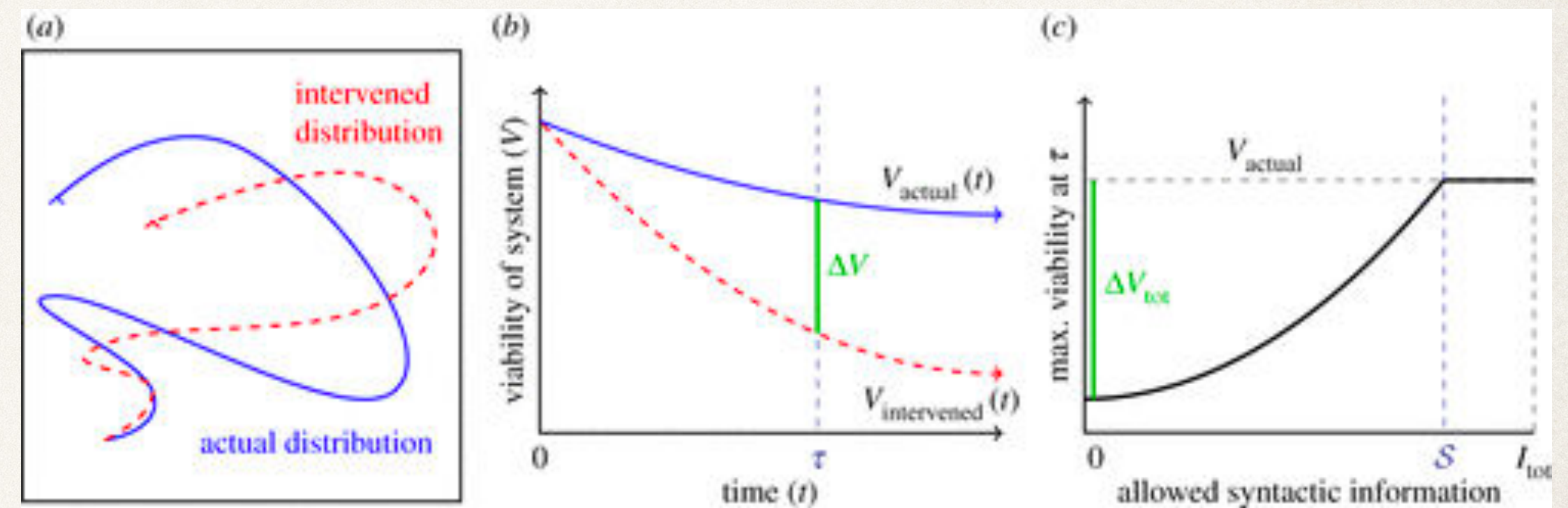
Semantic information - pros and cons

Strengths

- ❖ Goal-agnostic theory (we can swap “survival” with something else)
- ❖ Clarifying causal vs observational analysis

Limitations

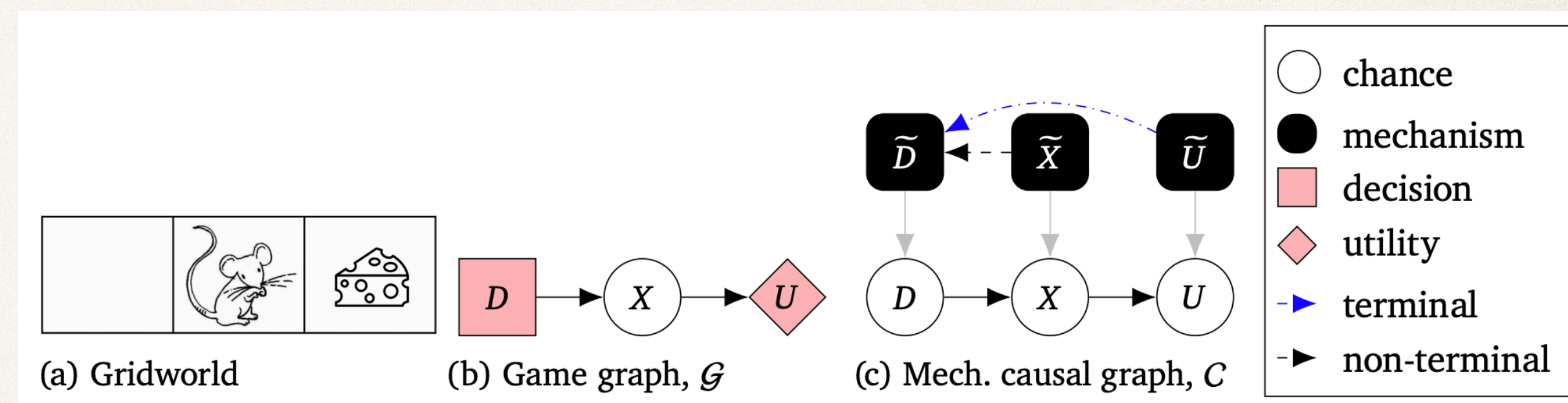
- ❖ Doesn't settle on a specific goal for agents
- ❖ Assuming we know the agent



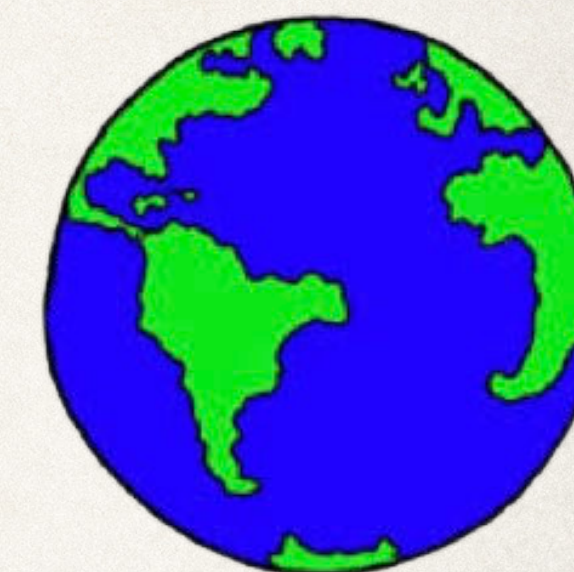
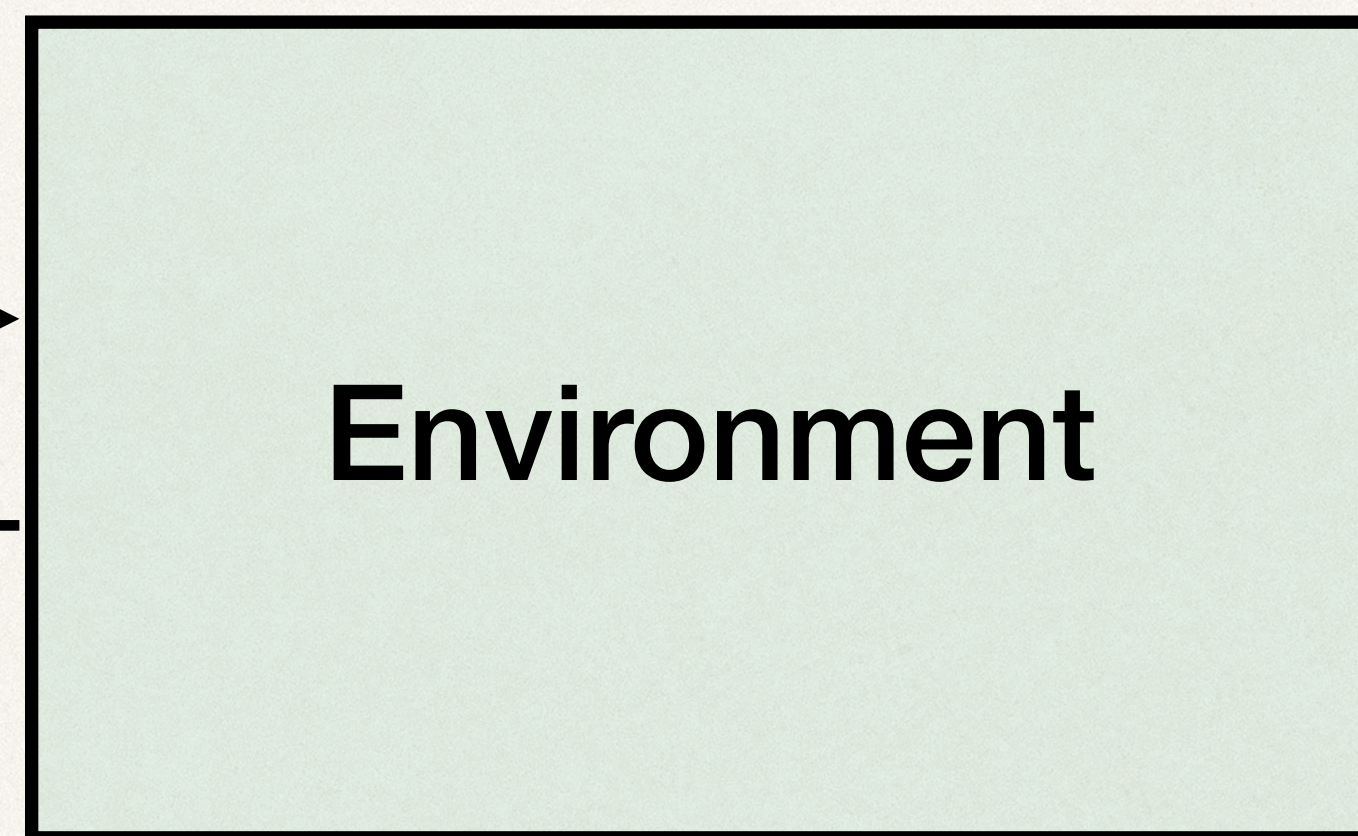
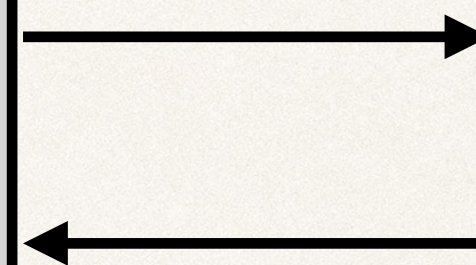
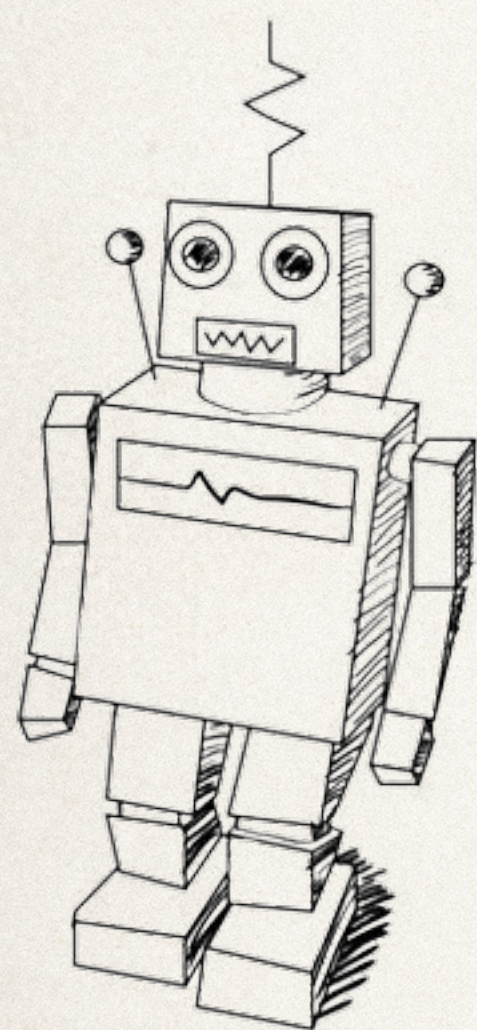
Kolchinsky, A., & Wolpert, D. H. (2018). Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface focus*, 8(6), 20180041.

Mechanised causal graphs

- ❖ Augment a graph reprinting decisions with “mechanisms”
- ❖ Mechanisms are parameters of “objective” variables
- ❖ Dependencies on mechanisms generally “reverse” causal chain (they need interventional data)



Kenton, Z., Kumar, R., Farquhar, S., Richens, J., MacDermott, M., & Everitt, T. (2022). Discovering Agents. arXiv preprint arXiv:2208.08345.



Mechanised causal graphs: “agents are systems that would adapt their policy if their actions influenced the world in a different way”

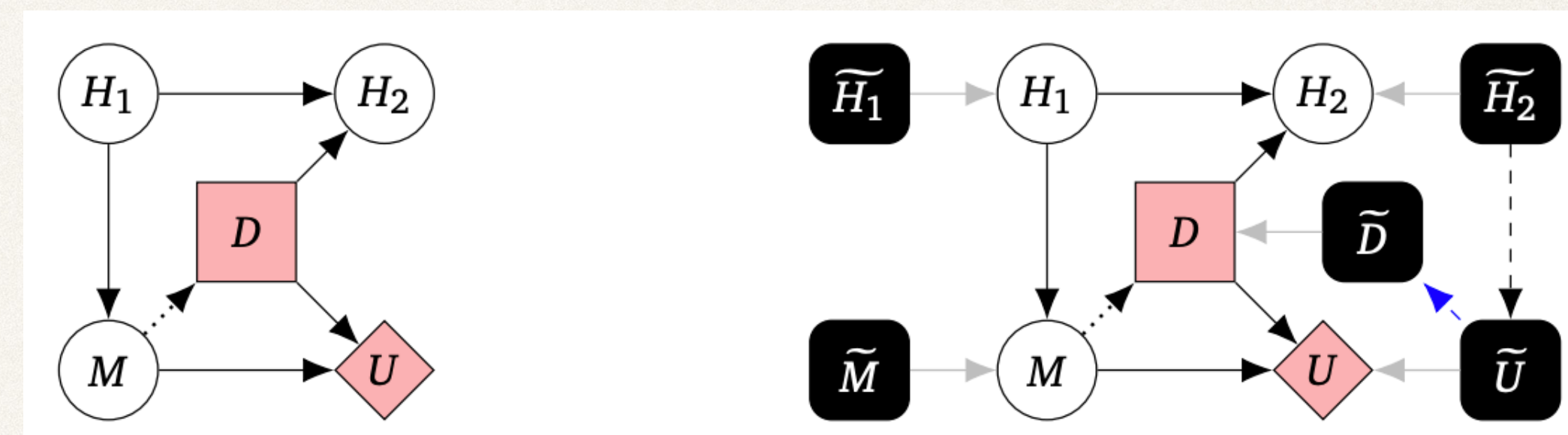
Mechanised causal graphs - pros and cons

Strengths

- ❖ Fully “Pearlian” description of agency
- ❖ Can pick up some interesting features of agency boundaries (what needs to be included and change given a certain mechanism, e.g., learning in RL)

Limitations

- ❖ Variables are chosen a-priori
- ❖ Interventions in the real world are difficult to obtain
- ❖ Are “soft” interventions (repeated observations) required for agency?



Kenton, Z., Kumar, R., Farquhar, S., Richens, J., MacDermott, M., & Everitt, T. (2022). Discovering Agents. arXiv preprint arXiv:2208.08345.

Causality-based methods

Advantages

- ❖ They take into account intrinsic notions of agency
- ❖ Pearl's causality is arguably the best account of causality we have
- ❖ Observer-independent

Disadvantages

- ❖ Is action necessarily related to causality?
- ❖ Choice of variables in causal models is somewhat subjective (cf. micro-state of a system)
- ❖ Actually, not really clear if these methods are observer-independent..

Relational methods

Agency with respect to something

Relational methods

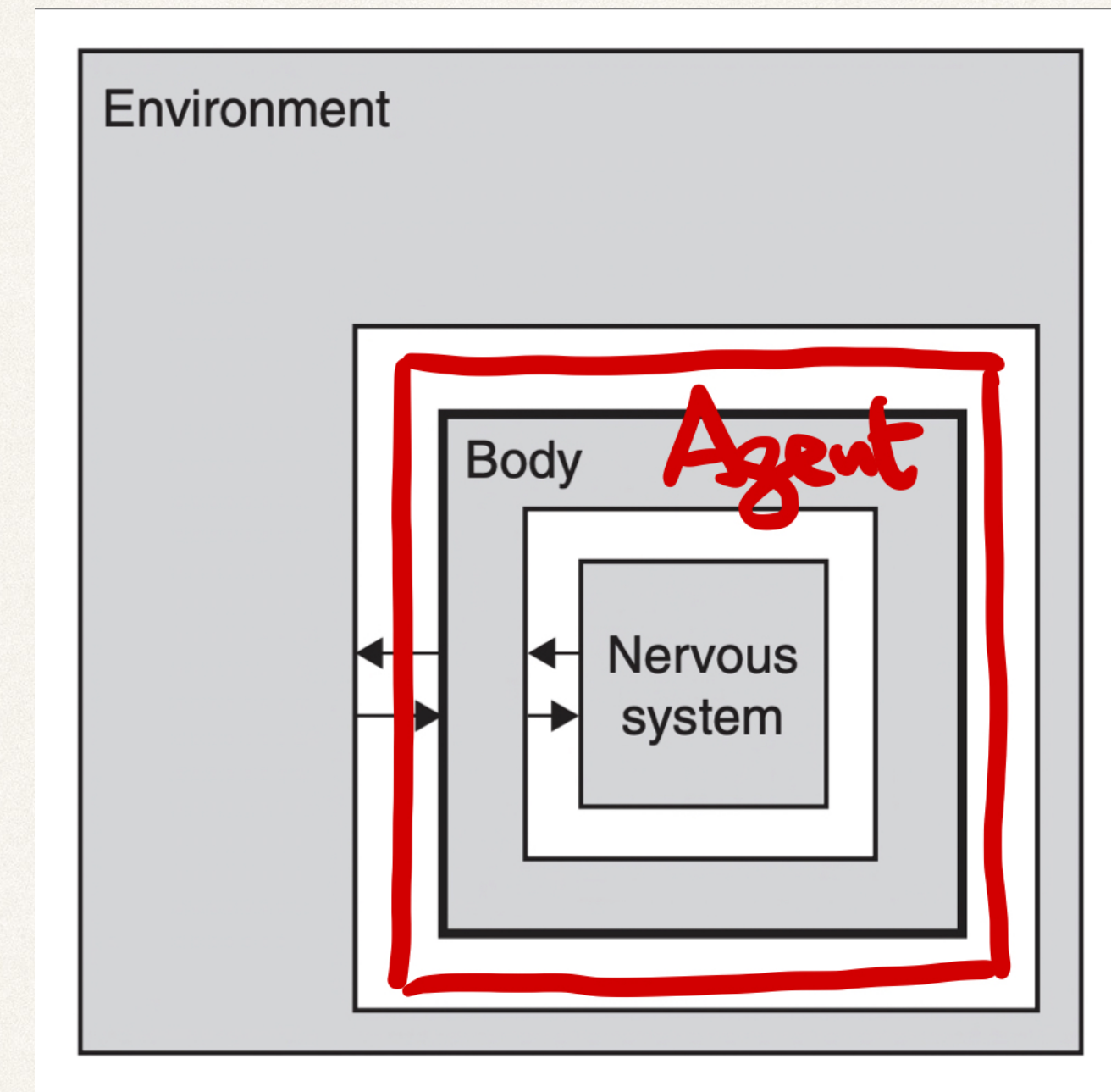
- ❖ **Main idea:** agency is in the way a system relates to other systems (environment, observer, other systems)
- ❖ **Inspiration:** cybernetics, Beer's work
- ❖ **Tools:** dynamical systems theory, systems theory, category theory, etc.

Examples:

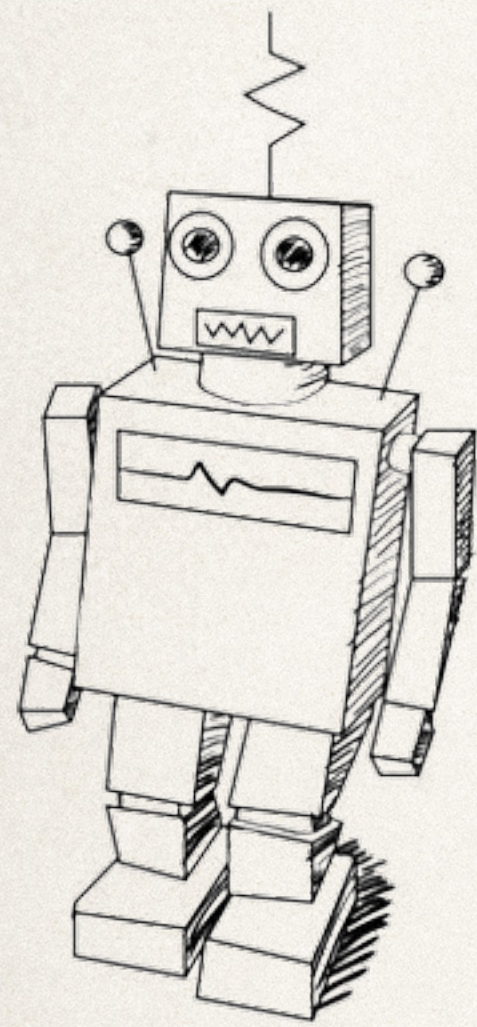
- ❖ Dynamical systems for agent-environment interactions
- ❖ Bayesian interpretation map
- ❖ Categorical agent-environment interactions

Dynamical systems for agent-environment interactions

- ❖ Take 3 dynamical systems and couple them
- ❖ Define adaptive fit: “An ~~animal~~ [agent] is adaptively fit to an environment only so long as it maintains its trajectory within this constraint volume [= the agent’s existence in state-space] despite the perturbations that it receives from its environment.”



Beer, R. D. (2008). The dynamics of brain–body–environment systems: A status report. *Handbook of Cognitive Science*, 99-120.



Beer's DSs: agents are systems that adaptively fit their environment

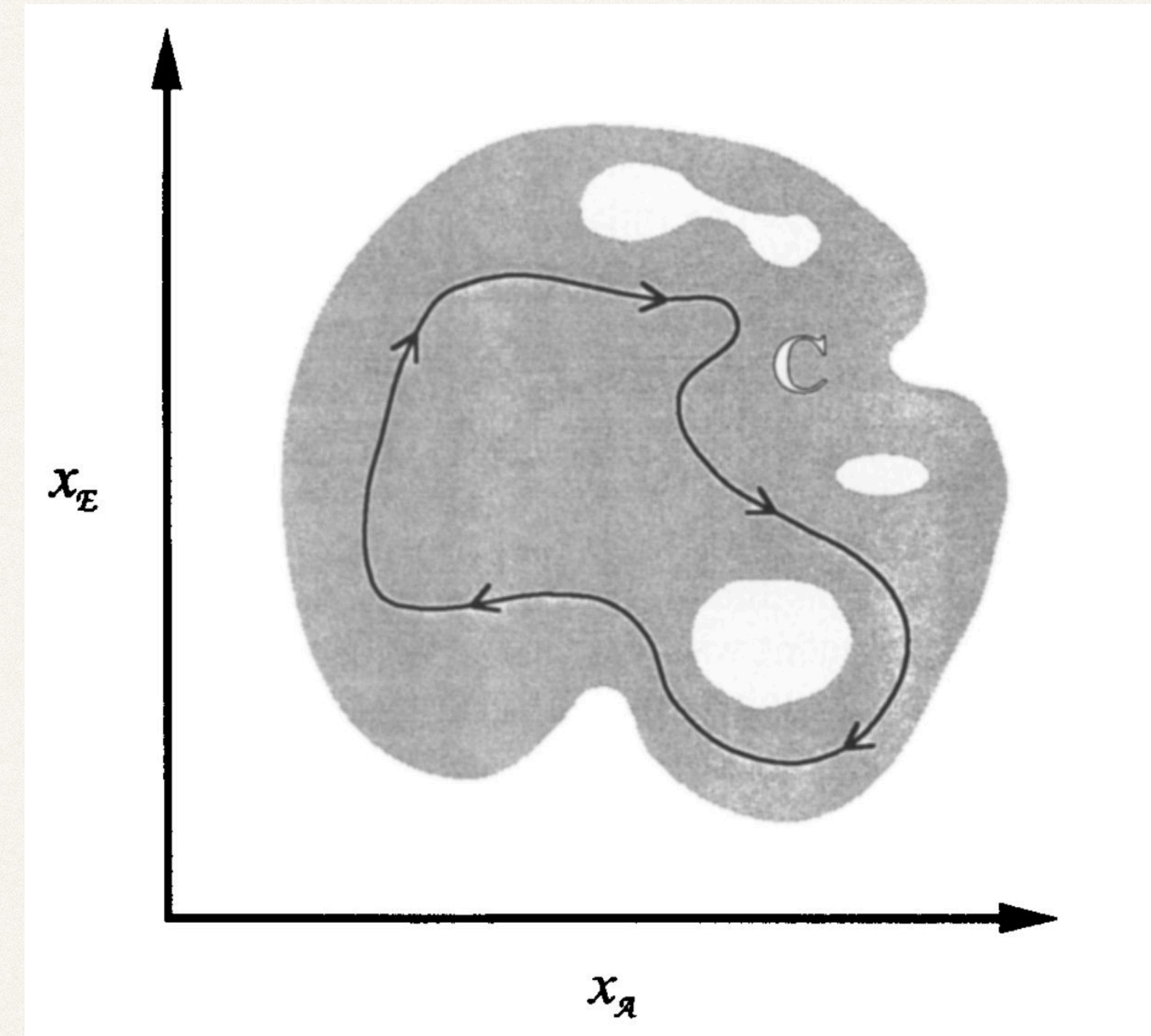
DSs for agent-environment interactions - pros and cons

Strengths

- ❖ Few assumptions
- ❖ Environment plays a role (adaptive fit)

Limitations

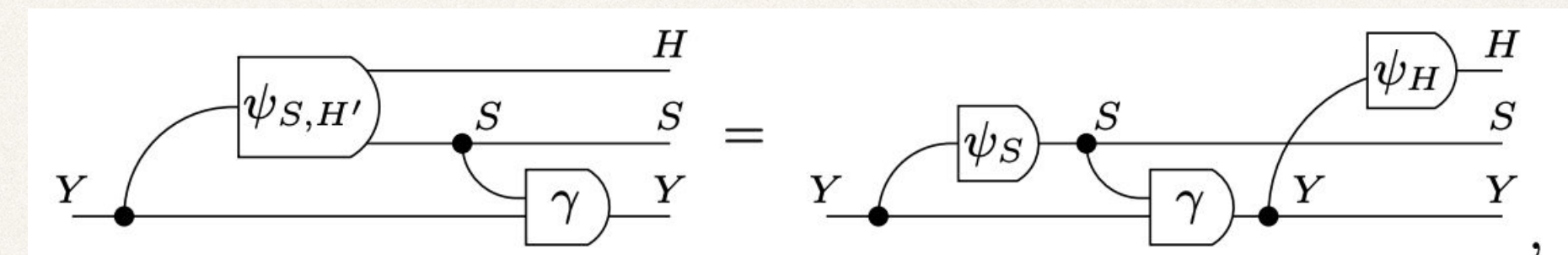
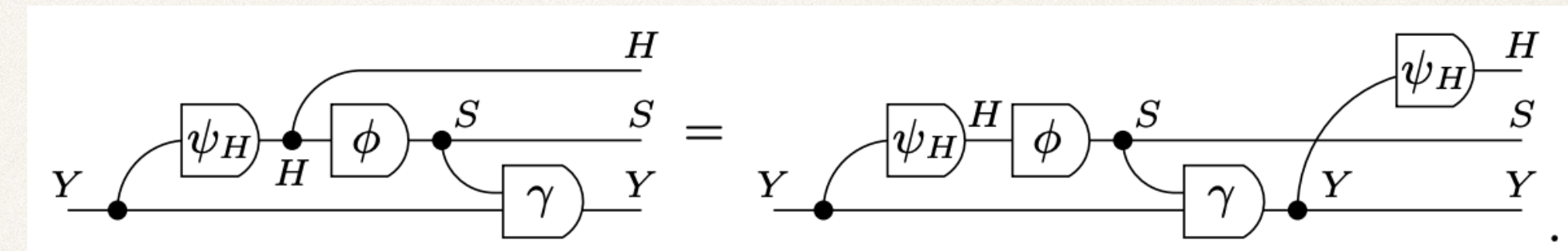
- ❖ “Adaptive fit” is never formalised
- ❖ Systems other than agents might show “adaptive fit”?
- ❖ Agents are assumed?



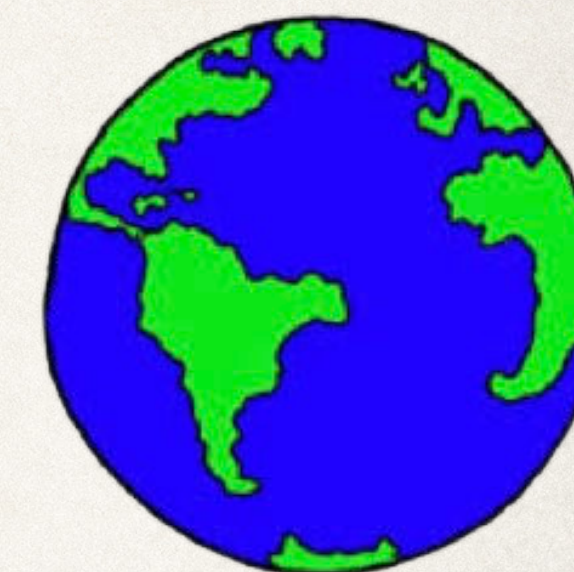
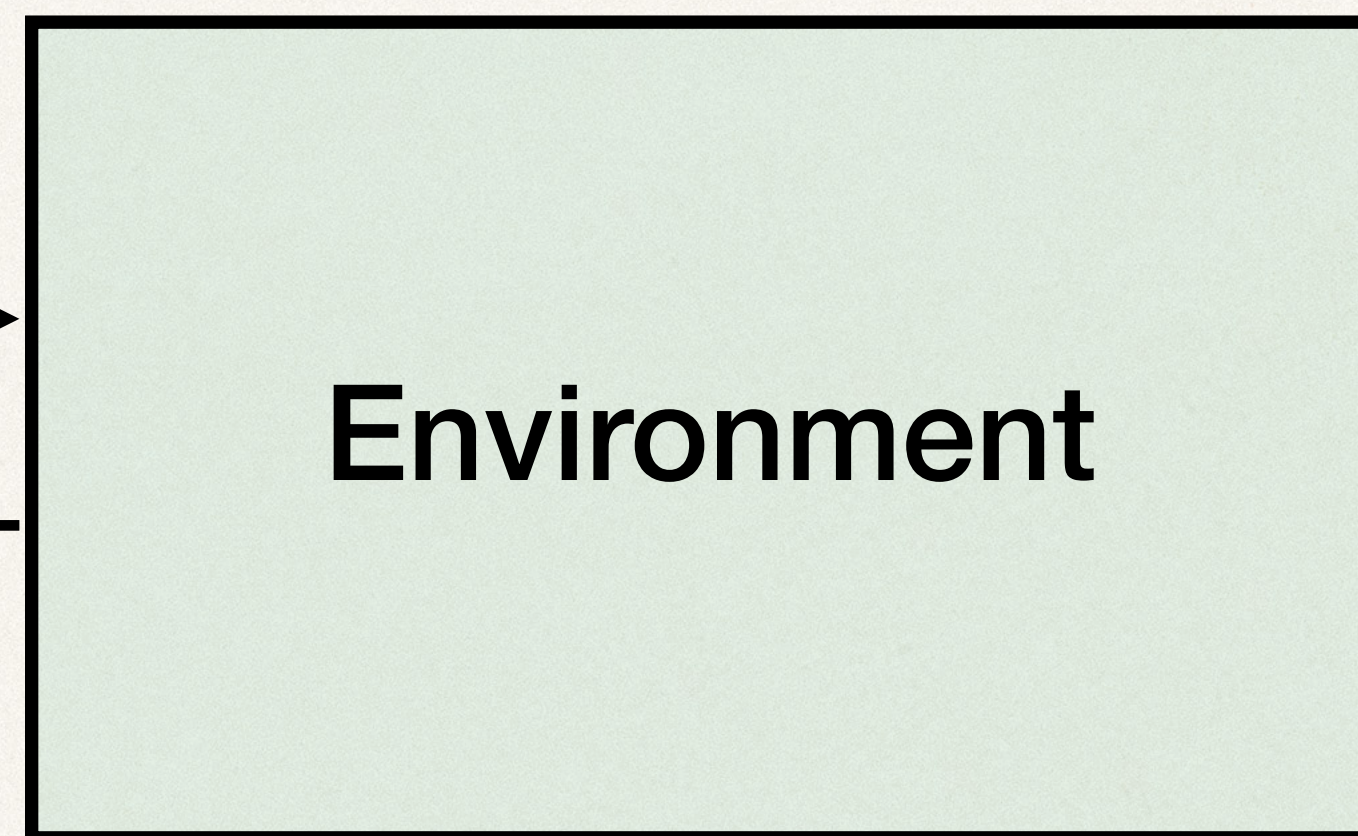
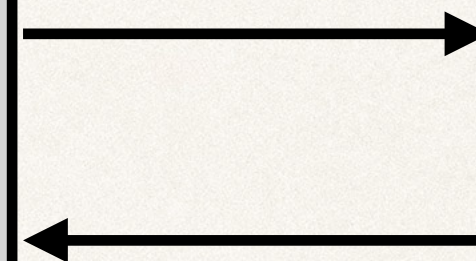
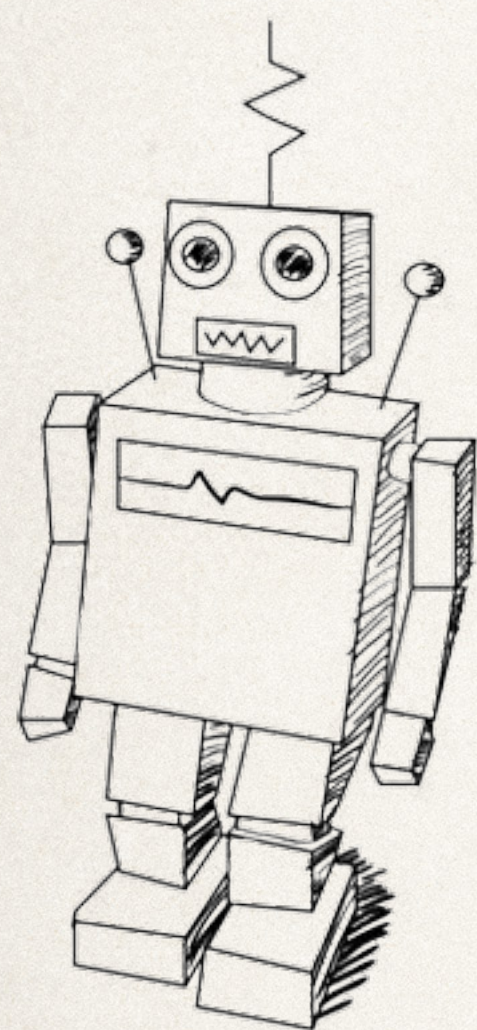
Beer, R. D. (1995). A dynamical systems perspective on agent-environment interaction. *Artificial intelligence*, 72(1-2), 173-215.

Bayesian interpretation map

- ❖ Take a dynamical system and build an “interpretation map”
- ❖ An interpretation map is a function that maps states of the system to “beliefs” as probability measures
- ❖ Perform Bayesian inference / filtering on these probabilities and check if this process is consistent with dynamical system evolution



Virgo, N., Biehl, M., & McGregor, S. (2022). Interpreting Dynamical Systems as Bayesian Reasoners. In Machine Learning and Principles and Practice of Knowledge Discovery in Databases

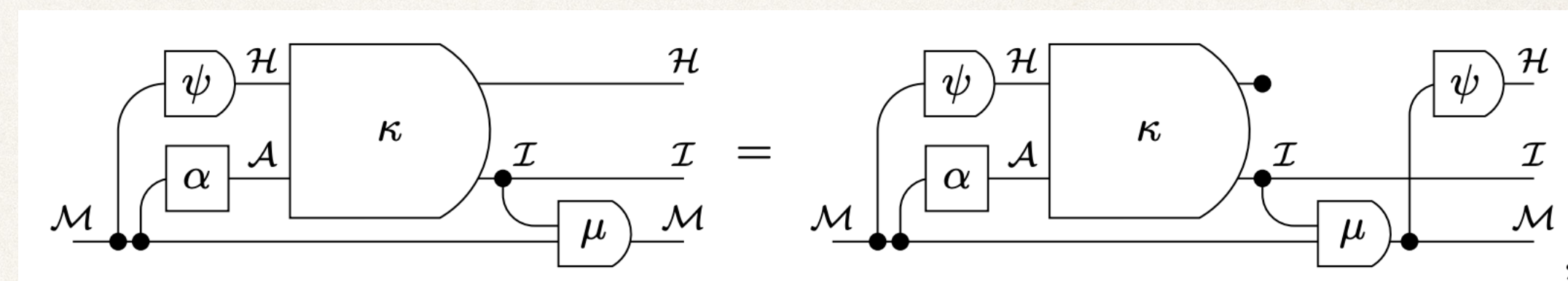


Bayesian interpretation map: if a system can be interpreted as performing Bayesian inference with some arbitrary model, then it's an agent

Bayesian interpretation map - pros and cons

Strengths

- ❖ Very general (category theory)
- ❖ Doesn't need to assume accurate knowledge of the environment (can fail causality test and still be an agent)



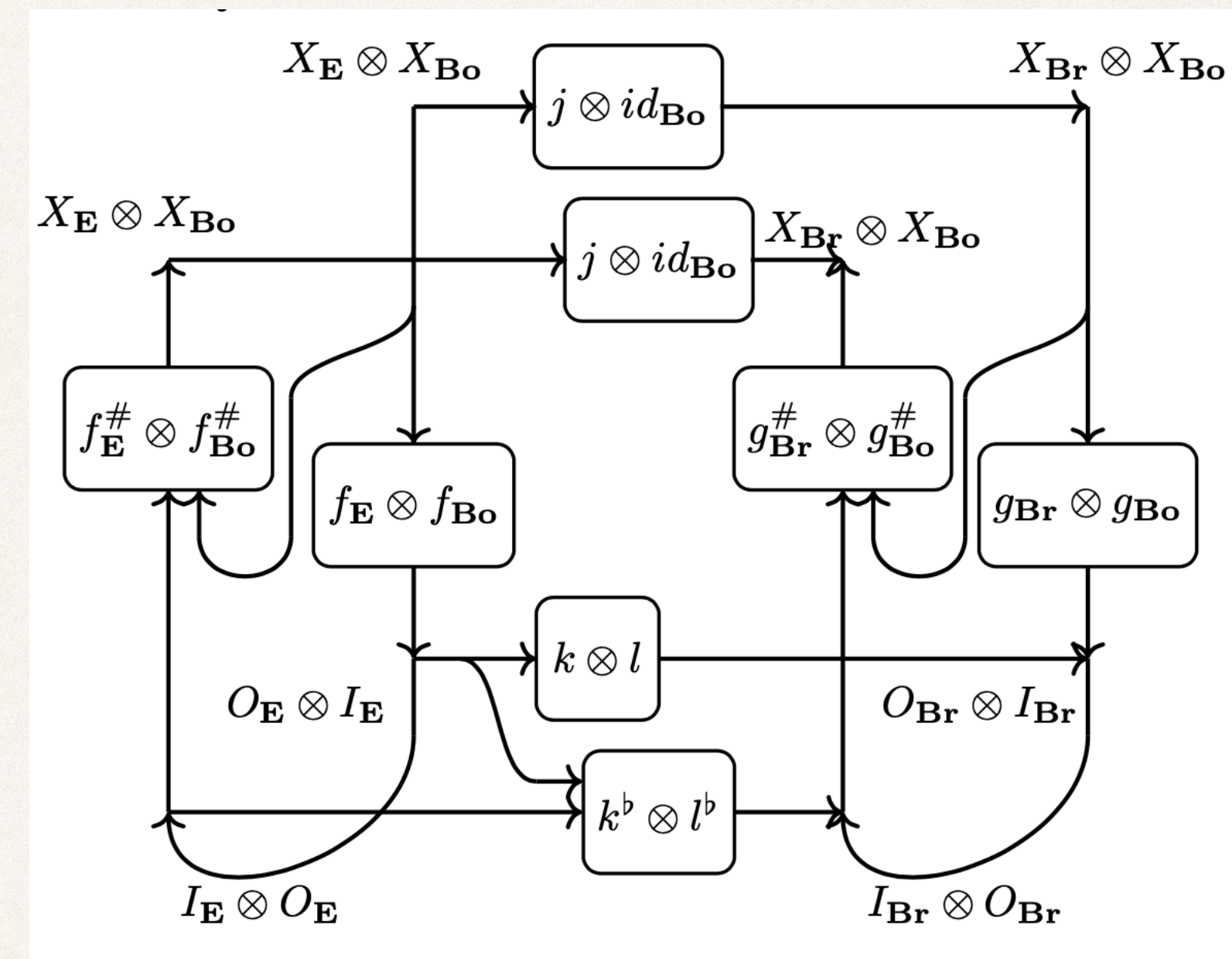
Limitations

- ❖ Doesn't consider real environment
- ❖ Including systems other than agents? (Thermostats, controllers, etc.)

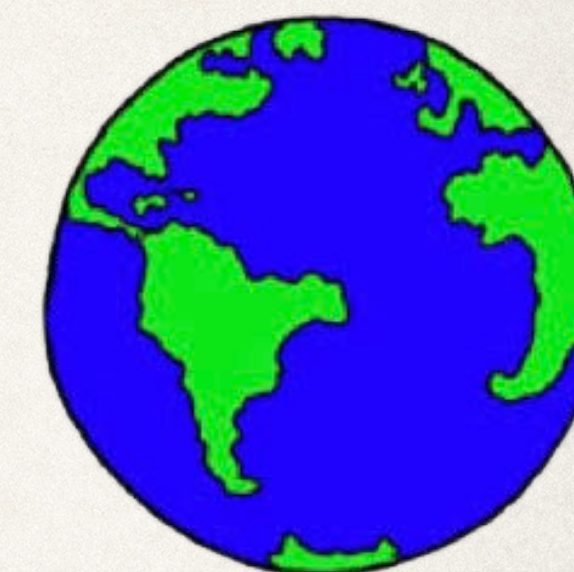
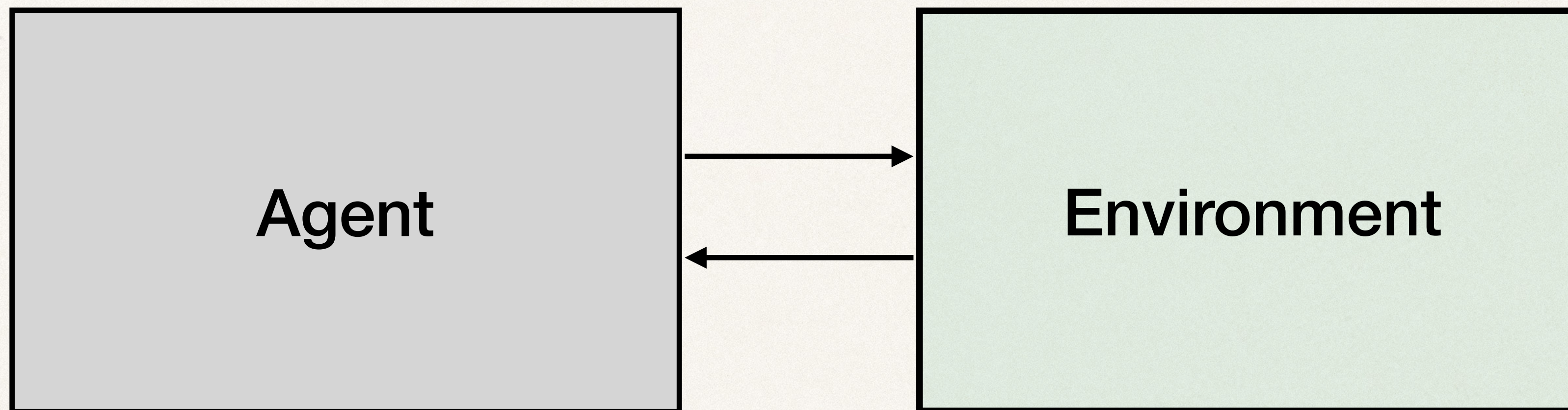
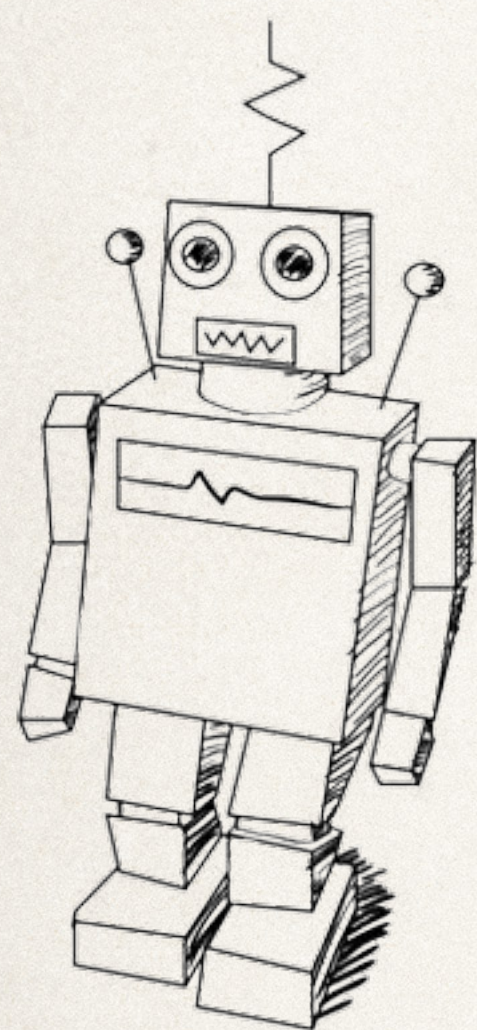
Biehl, M., & Virgo, N. (2022). Interpreting systems as solving POMDPs: a step towards a formal understanding of agency. *arXiv preprint arXiv:2209.01619*.

Categorical agent-environment interactions

- ❖ Take Beer's account, study physical connections among brain-body-environment
- ❖ Formalise “adaptive fit” using internal model principle from control theory (cf. law of requisite variety / good regulator theorem in cybernetics)
- ❖ Study “higher order” functional relations between brain and environment (via the body)



Baltieri, M. In progress.



Categorical agent-environment interactions: if there is a map between brain and environment while brain and environment are both physically connected to the body, we have a proto-agent

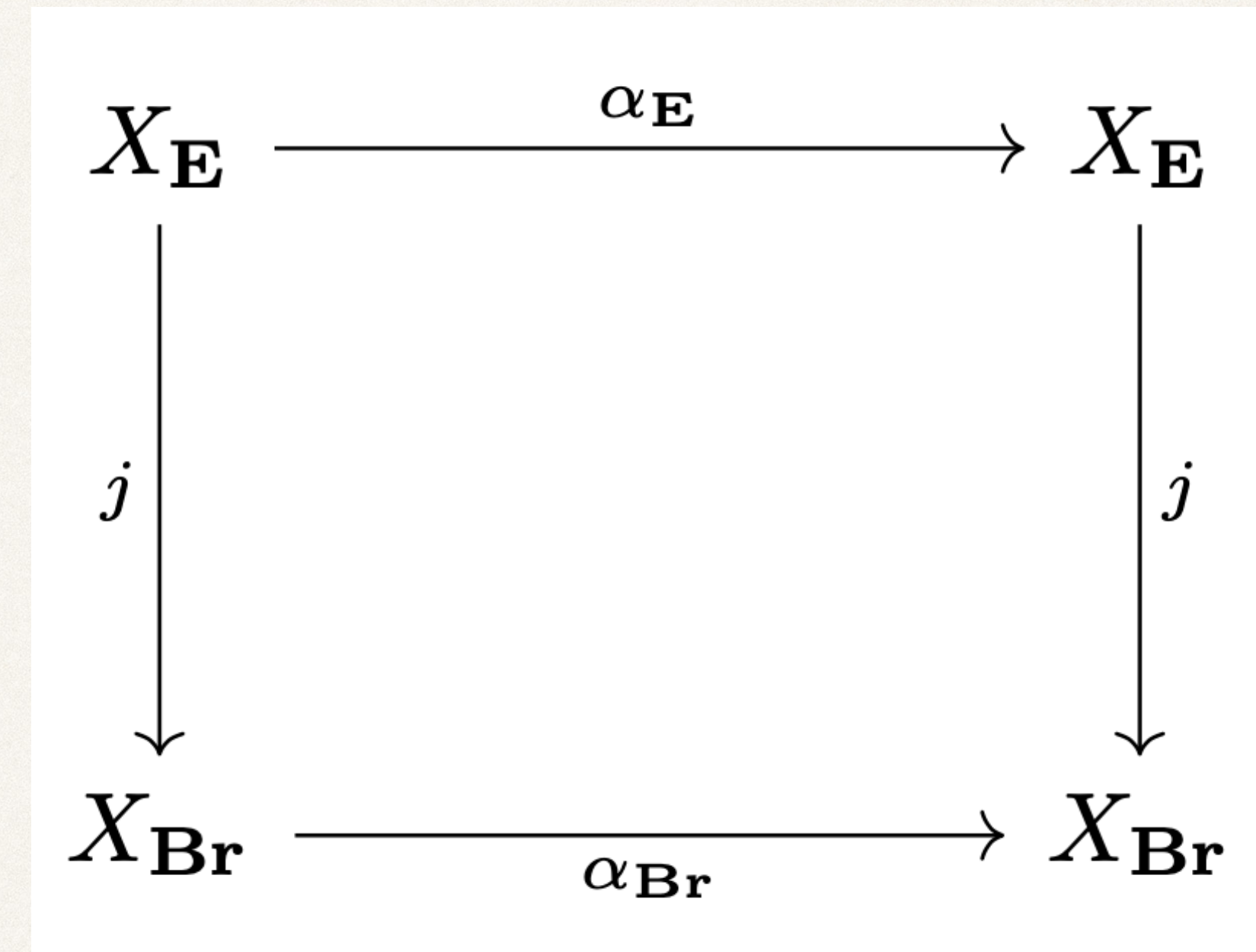
Categorical agent-environment interactions - pros and cons

Strengths

- ❖ Very general (category theory)
- ❖ It generalises FEP, Beer's proposal, probably other information-based ideas, partially Biehl + Virgo

Limitations

- ❖ Including systems other than agents?
- ❖ No account of causality
- ❖ Limited to "real" environments unlike Biehl + Virgo



Baltieri, M. In progress.

Relational methods

Advantages

- ❖ Most general domain of applications (physical vs non-physical, sets vs. graphs vs. probabilities, etc.)
- ❖ Agency as a relational property, we can in principle add other agents, observers, etc. and account for how they affect agency
- ❖ Few assumptions (due to their generality)

Disadvantages

- ❖ Too general to say something practically useful?
- ❖ Not obvious how causal claims could be considered in this class of approaches
- ❖ Hard to know if we are capturing specifically a notion of agency or something else (maybe related to it)

Conclusion

- ❖ Prediction-based methods: agency in the eye of the beholder
- ❖ Causality-based methods: agency as a property intrinsic to a system
- ❖ Relational methods: agency with respect to something

What should your definition of agency include?

(Can you define agency?)